

An Integrated Optimization Model for Home Healthcare Routing and Scheduling with Synchronization, Break Scheduling, and Temporal Stability

Dorsa Abdolhamidi^{a,1}, Virginie Lurkin^a

^a*University of Lausanne, HEC Lausanne Faculty of Business and Economics, Quartier de Chamberonne, Lausanne, 1015, Vaud, Switzerland*

Abstract

Home healthcare organizations must construct daily schedules that use constrained caregiver capacity efficiently while preserving continuity of care, predictable appointment times, household-level coordination, and compliance with working-time regulations. This paper develops an integrated optimization framework for the home healthcare routing and scheduling problem that incorporates synchronized visits, patient-caregiver preferences and previous-caregiver continuity, coordination of co-located patients, location-specific and location-flexible breaks, and appointment-time stability. The problem is solved on a daily basis, with historical information entering through two temporal references: the appointment time most recently communicated to the patient and the baseline routine established at the beginning of the care episode. This dual-horizon formulation distinguishes the cost of changing an announced schedule from the cost of drifting away from an established routine. We compare a mixed-integer linear formulation, a solver-native list-and-interval formulation (Hexaly), and two heuristic benchmarks. The computational study uses real operational data from a partner home healthcare organization, with limited controlled modification introduced to activate break and synchronization features not exercised in this particular partner's dataset. The results show that coverage depends not only on aggregate caregiver availability but also on the temporal, spatial, and qualification alignment of capacity with demand; on these instances, where requested workload plus estimated average travel typically exceeds contracted capacity, the model serves a share of demand comparable to the organization's own realized schedules, and increasing the priority on coverage does not close the shortfall. They further show that moderate changes in managerial priorities can substantially reduce the modeled penalties associated with caregiver continuity and appointment-time stability, with limited additional travel, waiting, and overtime.

Keywords: Home healthcare routing and scheduling; Patient-centered planning; Continuity of care; Appointment-time stability; Mixed-integer linear programming

1. Introduction

Population aging is increasing the demand for long-term care. At the same time, the rising cost of institutional care and patients' preference to remain at home are encouraging a shift toward home healthcare (HHC) services. By 2030, one in six people worldwide will be aged 60 or older, compared with one in eleven in 2020 (World Health Organization, 2024). The demographic shift is

Email addresses: dorsa.abdolhamidi@unil.ch (Dorsa Abdolhamidi), virginie.lurkin@unil.ch (Virginie Lurkin)

¹Corresponding author

particularly marked in Europe, where the population aged 65 and older is projected to increase by 43% between 2019 and 2050 (Eurostat, 2020).

Meeting this growing demand is difficult because the workforce available to provide care is increasingly constrained. Switzerland illustrates this pressure. The share of residents aged 65 and older, currently around 19%, is projected to reach 26% by 2055 (Federal Statistical Office (FSO), 2025), while the number of working-age individuals per retiree is expected to fall from approximately four today to two by mid-century (Alonso and Grampp, 2025). Under these conditions, daily planning involves more than assigning visits to available caregivers or minimizing travel. Planners must use limited capacity effectively while preserving continuity of care, reliable appointment times, and acceptable working conditions.

Available working time is not necessarily usable care capacity. A caregiver may have idle time in a schedule and still be unable to serve an additional appointment because the remaining time occurs in the wrong place, falls outside the patient’s time window, or belongs to a shift without the required qualifications. Travel times, synchronized services, and mandatory breaks further restrict which appointments can be added to a route. Coverage, operational efficiency, and service quality are therefore closely connected. The planning problem is not only to construct a feasible schedule, but also to understand what operational costs arise when greater importance is placed on continuity, timing stability, or other patient-centered criteria.

This paper studies a daily routing and scheduling problem arising in the planning environment of a Swiss software provider that develops solutions for HHC organizations. The operational data used in the study were supplied, under a data-sharing agreement, by one of its care-delivery partners. The planner must assign appointments to caregiver shifts, determine the sequence of visits, and schedule each service within its prescribed time window. Assignments must respect caregiver qualifications. Patient–caregiver preferences and continuity with previously assigned caregivers are represented as distinct service criteria (Braekers et al., 2016). Some services require several caregivers to arrive simultaneously (Bredström and Rönnqvist, 2008; de Aguiar et al., 2023), while visits to patients living at the same address may need to occur close together. Caregiver shifts may also include mandatory breaks, taken either at a specified location or flexibly along the route (Rest and Hirsch, 2016; Goel and Vidal, 2013). Finally, recurring appointments may need to remain close both to the time most recently communicated to the patient and to the routine established earlier in the care episode.

Each operating day is solved separately. Previously communicated appointment times, base-line routines, and previous caregiver assignments enter the model as fixed historical inputs rather than as decisions in a multi-period optimization problem. The underlying instances contain real appointments, caregiver shifts, qualifications, locations, time windows, and travel information. Of the features in the model, five are present natively in the partner’s records: patient–caregiver preferences, continuity with the previous caregiver, household co-location, and both the short- and long-term appointment-time references. Two further features, mandatory breaks and synchronized visits, are used elsewhere in the software provider’s client base but were not exercised by this particular partner during the study month. We therefore introduce limited, controlled modifications to activate them, so that the complete model is exercised on operationally grounded instances. None of the managerial findings reported in Section 5 depends on these modifications: each is reproduced on the unmodified instances, as reported in Appendix E. The modifications are consequently not used as the basis for any conclusion specific to the data-providing organization.

Several planning criteria must be considered at the same time. Greater continuity may require additional travel. Keeping an appointment close to the time most recently communicated to the patient may move it away from the patient’s usual routine. Unused caregiver time may remain unavailable for additional visits because it is poorly aligned with the timing, location, or qualifica-

tion requirements of demand. These interactions call for a model that can do more than produce a low-cost route. It should also allow planners to examine how improvements in one criterion affect the others.

The home healthcare routing and scheduling problem (HHCSP) literature has addressed many of the individual features involved. Existing surveys document substantial work on routing, scheduling, synchronization, caregiver assignment, and service consistency in HHC (Fikar and Hirsch, 2017; Cissé et al., 2017; Grieco et al., 2021; Euchí et al., 2022; Masmoudi et al., 2024). Most studies, however, consider selected subsets of these requirements. Temporal consistency is also commonly represented through hard constraints or discrete service patterns. These approaches are appropriate when a prescribed level of consistency must be guaranteed, but they provide less information about the operational cost of gradually improving schedule stability.

Two gaps motivate this study. First, to the best of our knowledge, existing HHCSP models do not jointly represent synchronized visits, preference-based continuity, continuity with the previous caregiver, household-level temporal coordination, location-specific and location-flexible breaks, and appointment-time stability within a single daily planning model. Second, continuous-penalty formulations from the consistent vehicle routing literature (Kovacs et al., 2014a, 2015) have not been applied to HHC in a way that distinguishes the most recently communicated appointment time from the longer-term baseline routine; two references that a schedule may preserve independently.

The paper makes four contributions. First, we formulate an integrated mixed-integer linear programming (MILP) model for daily HHC routing and scheduling that jointly represents the features above within a single daily model. Second, we introduce a dual-horizon formulation that penalizes deviations from the most recently communicated appointment time separately from deviations from the patient's baseline routine. This makes it possible to distinguish the cost of changing an announced schedule from the cost of drifting away from an established routine. Third, we compare a classical MILP formulation solved with Gurobi, a list-and-interval formulation implemented in Hexaly, and two independently implemented heuristic benchmarks. Fourth, using operational data from a partner HHC organization, we examine the relationships among coverage, operational efficiency, patient-caregiver preferences, caregiver continuity, and short- and long-term appointment-time stability.

The computational study examines whether the proposed model can produce reliable schedules within the time available for daily planning, and whether the main operational findings depend on a particular solution method or implementation.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the mathematical formulation and the Hexaly representation. Section 4 describes the computational study. Section 5 discusses the managerial results, and Section 6 concludes the paper.

2. Literature review

HHCSP combines caregiver assignment, visit sequencing, and service-time decisions with requirements such as qualifications, time windows, continuity of care, and service consistency. Recent surveys document the breadth of this literature (Fikar and Hirsch, 2017; Grieco et al., 2021; Euchí et al., 2022; Masmoudi et al., 2024; Fu et al., 2025). Many studies focus on selected combinations of these requirements, whereas the planning setting considered here requires several operational and patient-related criteria to be represented jointly. We review three streams that are most relevant to this paper: integrated HHC routing and scheduling, service consistency in vehicle routing, and continuity and temporal requirements in HHC.

2.1. Integrated routing and scheduling with synchronization

Integrated HHCRSP models determine caregiver assignments, visit sequences, and service times within a single planning problem. An early contribution is [Rasmussen et al. \(2012\)](#), who develop a branch-and-price method for a problem combining routing, scheduling, synchronization, and preference-weighted caregiver assignment. [Naderi et al. \(2023\)](#) extend this line of work through a logic-based Benders decomposition for an uncertain setting.

Synchronization is important when a service requires several caregivers to attend the same appointment simultaneously. This requirement was formalized by [Bredström and Rönnqvist \(2008\)](#) and has since been combined with other operational features in several HHCRSP variants. [de Aguiar et al. \(2023\)](#) consider synchronized services and team coordination. [Liu et al. \(2021\)](#) combine synchronization with time windows, caregiver preferences, lunch breaks, and flexible departure decisions, while [Wang et al. \(2025\)](#) study synchronization, lunch breaks, and coordination across several service centers. Other extensions incorporate continuity and rostering ([Algendi et al. 2025](#)), precedence relations between visits ([Kummer et al. 2024](#)), or compact constraint-programming formulations ([Montemanni et al. 2025](#)).

Taken together, these studies show that synchronization can be modeled alongside qualifications, time windows, preferences, breaks, and, in some cases, continuity or rostering. They do not, however, directly combine these operational requirements with continuous appointment-time penalties measured against both a recently communicated schedule and a longer-term patient routine.

A wide range of solution methods has been proposed for integrated HHCRSP models. Exact approaches include branch-and-price ([Rasmussen et al. 2012](#)) and logic-based Benders decomposition ([Naderi et al. 2023](#)). Metaheuristics and matheuristics include population-based methods ([Decerle et al. 2018](#); [Masmoudi et al. 2023](#); [Kummer et al. 2024](#)), multi-neighborhood local search ([Ceschia et al. 2026](#)), and adaptive large neighborhood search ([Wang et al. 2025](#); [Bonomi et al. 2025](#)). This literature provides the computational basis for solving rich HHC models at operational scale.

2.2. Service consistency in vehicle routing

The temporal-stability component of the present study is related to the consistent vehicle routing problem (ConVRP). [Kovacs et al. \(2014b\)](#) distinguish arrival-time consistency from person-oriented consistency. The former concerns serving a customer at approximately the same time across repeated visits, while the latter concerns repeated service by the same driver or a limited set of drivers. The ConVRP was introduced by [Groër et al. \(2008\)](#) and further developed by [Tarantilis et al. \(2012\)](#), who use route templates to limit variation in customer service times.

The closest methodological precedent for this paper is the Generalized ConVRP of [Kovacs et al. \(2014a\)](#), in which arrival-time variation is penalized in the objective rather than controlled only through hard constraints. [Kovacs et al. \(2015\)](#) extend this approach to a multi-objective setting. Related studies examine the interaction between service consistency and workload balance ([Mancini et al. 2021](#)), demand variation ([Luo et al. 2015](#)), and heterogeneous fleets ([Stavropoulou 2022](#)).

These studies provide useful modeling ideas, but HHC introduces requirements that are generally absent from standard vehicle-routing models, including clinical qualifications, patient–caregiver compatibility, synchronized services, and working-time constraints. Existing consistency models also generally measure variation around one temporal reference or across realized arrival times. In the setting considered here, two references are relevant: the appointment time most recently communicated to the patient and the baseline routine recorded earlier in the care episode. The first represents a recent schedule commitment, whereas the second is used to measure longer-term schedule drift.

2.3. Patient-related temporal features in home healthcare

Patient-related HHC planning includes both assignment preferences and continuity across repeated visits. Assignment costs may represent preferences concerning individual caregivers, language, gender, or other service attributes. Such preferences are included through explicit assignment-cost matrices (Liu et al., 2021; Wang et al., 2025; Yadav and Tanksale, 2022) or preference-weighted caregiver assignment (Rasmussen et al., 2012).

Continuity across visits is a related but distinct consideration. It may be represented by penalizing reassignment from a caregiver who previously served the patient. This approach appears in rolling-horizon (Güven-Kocak et al., 2024; Grenouilleau et al., 2019), robust (Carello et al., 2018), large-neighborhood (Bonomi et al., 2025), and rostering-integrated formulations (Algendi et al., 2025). Other studies limit the number of caregivers assigned to each patient or the number of patients assigned to each caregiver (Cappanera and Scutellà, 2022; Delaet et al., 2024; Fathollahi-Fard et al., 2021). These formulations capture related aspects of caregiver consistency, but they do not necessarily distinguish patient-caregiver preferences from continuity with the caregiver who served the patient most recently.

Appointment-time stability is another form of service consistency. Home-care recipients, particularly elderly and chronically ill patients, may be sensitive to schedule changes (Cappanera and Scutellà, 2022), and schedule inconsistency has been associated with patient complaints (Mommcheva et al., 2025). Existing HHC models represent temporal stability through structural planning concepts (Mankowska et al., 2014; Grenouilleau et al., 2019), hard constraints (Demirbilek et al., 2019), or discrete pattern-based formulations (Cappanera and Scutellà, 2022; Bonomi et al., 2025). These approaches are appropriate when a prescribed level of stability must be enforced, but they are less direct for measuring gradual trade-offs between operational cost and appointment-time consistency.

Household-level coordination concerns a different temporal relationship. When several patients live at the same address, planners may wish to schedule their visits close together to coordinate access to the home or reduce disruption to the household. Synchronization constraints coordinate several caregivers attending the same appointment, while precedence constraints order multiple services for the same patient (Kummer et al., 2024). Neither mechanism directly represents a soft penalty on the temporal distance between visits to distinct patients at the same address.

Break scheduling must also be considered when shift duration and the applicable working-time rules require caregivers to take rest breaks during the day. The broader vehicle-routing literature includes models for driver breaks (Coelho et al., 2016; Goel and Vidal, 2013), but breaks are not represented in many HHC formulations. Among the studies that include them, Rest and Hirsch (2016) consider flexible break insertion, whereas Liu et al. (2021) and Wang et al. (2025) model breaks associated with designated locations. The joint treatment of location-specific and location-flexible breaks in an integrated HHC model remains limited.

Table 1 summarizes the studies most closely related to the feature dimensions considered in this paper. It is not intended as a systematic review or a ranking of contributions. Its purpose is to show which combinations of operational and temporal requirements have been represented in previous models.

2.4. Research gaps and positioning of this work

The literature provides established formulations for caregiver assignment, routing, synchronization, breaks, and several forms of service consistency. These elements are generally studied in selected combinations. To the best of our knowledge, no existing daily HHCRSP model jointly represents synchronized services, patient-caregiver assignment preferences, continuity with the

Table 1: Comparison of related work along the feature dimensions of the proposed integrated model. ✓: feature addressed in full; ●: feature addressed in a limited or partial form. *Sync.*: synchronized visits; *Pref.*: soft patient-caregiver assignment; *Prev. caregiver*: continuity of care from the previous visit; *Household*: penalized continuous temporal proximity between visits to co-located patients; *Loc.-spec. break*: mandatory break at a designated location; *Loc.-flex. break*: mandatory break that can be inserted along the route; *Short-term stab.*: control over deviations from the appointment time most recently communicated for the patient; *Long-term stab.*: control over deviations from a baseline appointment time established at the start of the care episode; *Cont. temporal penalty*: continuous-valued penalty in the objective rather than hard constraint or discrete pattern; *Real data*: validation on operational data from a partner organization.

Reference	Sync.	Pref.	Prev. caregiver	Hh. coord.	Loc.-spec. break	Loc.-flex. break	Short-term stab.	Long-term stab.	Cont. penalty	Real data
<i>Integrated HHC routing & scheduling</i>										
Rasmussen et al. (2012)		✓								
Naderi et al. (2023)	✓		✓							
de Aguiar et al. (2023)	✓									✓
Liu et al. (2021)	✓	✓			✓					●
Wang et al. (2025)	✓	✓			✓					
Algendi et al. (2025)	✓		✓		●					✓
Montemanni et al. (2025)	✓									
Decerle et al. (2018, 2019)	✓									✓
Kummer et al. (2024)	✓									✓
Yadav and Tanksale (2022)	✓	✓			●					✓
Delaet et al. (2024)										✓
<i>HHC consistency-focused</i>										
Cappanera and Scutellà (2022)							✓ [†]			
Güven-Kocak et al. (2024)			✓							✓
Grenouilleau et al. (2019)			✓				●			✓
Carello et al. (2018)			✓							✓
Bonomi et al. (2025)			✓				●			✓
Mankowska et al. (2014)	✓						●			
Demirbilek et al. (2019) [§]							✓ [*]			
Rest and Hirsch (2016)						✓				✓
<i>ConVRP precedent</i>										
Kovacs et al. (2014a)							✓		✓	
Kovacs et al. (2015)							✓		✓	
This paper	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

[§] Demirbilek et al. (2019) consider a single-nurse setting in which continuity of care is trivially preserved by the problem structure.

^{*} Short-term stability enforced as a hard constraint (same time of the day), not as a penalized continuous objective.

[†] Short-term stability modeled through discrete time-slot patterns, not as a continuous deviation penalty.

previous caregiver, soft coordination of visits to co-located patients, both location-specific and location-flexible breaks, and appointment-time stability.

The consistency literature also points to a more specific modeling gap. Continuous penalties have been used in vehicle routing to control arrival-time variation, while HHC studies have generally represented temporal stability through hard requirements, planning patterns, or a single temporal reference. In the setting considered here, two references are relevant: the appointment time most recently communicated to the patient and the baseline routine recorded earlier in the care episode. The former represents a recent commitment, whereas the latter provides a reference for measuring longer-term schedule drift.

We address these gaps through an integrated daily assignment, routing, and scheduling model. Historical appointment times and caregiver assignments enter each daily problem as fixed inputs. The computational study uses real operational data supplied by a partner HHC organization under a data-sharing agreement with the Swiss software provider involved in the study. Limited controlled modifications are used to activate break and synchronization features that were absent from this particular dataset. The analysis examines the trade-offs among coverage, operational cost, patient-caregiver assignment preferences, caregiver continuity, household coordination, and short- and long-term appointment-time stability.

3. Mathematical formulation

This section formalizes the daily planning problem faced by the HHC provider. The formulation integrates assignment, routing, and scheduling decisions while representing the operational, regulatory, and patient-centered requirements that shape usable care capacity in practice. In doing so, it provides the basis for the trade-off analyses developed later in the paper, where routing efficiency, coverage, continuity of care, temporal stability, and other service-quality criteria are evaluated jointly.

Section 3.1 defines the problem setting and notation. Section 3.2 presents the mixed-integer linear programming formulation used to represent the integrated decision problem. Section 3.3 then gives an equivalent formulation in the Hexaly modeling language. This second representation is included to make the commercial-solver benchmark transparent and reproducible, and to compare the classical MILP formulation with a solver-native list- and interval-based representation in the computational study of Section 4.

3.1. Problem description and notation

We consider the day-ahead planning problem of an HHC provider. For each operating day, a set of patient appointments must be assigned to caregiver shifts, sequenced into routes, and scheduled within prescribed time windows. Historical information from previous operating days is treated as fixed input to the daily problem: previously communicated appointment times, baseline patient routines, and previously assigned caregivers enter the model as reference parameters rather than as decisions in a coupled multi-period program. This modeling choice reflects the provider's planning process, in which daily schedules are constructed against an existing operational history.

Each caregiver shift originates and terminates at designated locations, operates within a contracted working period, and is associated with a transportation mode (*e.g.*, car, bicycle, public transit, or walking). Travel times are therefore shift-dependent: the parameter τ_{ijk} denotes the travel time from node i to node j when the trip is performed by shift k . Appointments may require specific caregiver qualifications, and some appointments require synchronized service by multiple caregivers arriving simultaneously.

The formulation represents the operational and patient-centered requirements that determine whether nominal capacity can be used in practice. Shifts may include mandatory breaks, which can be location-specific (*e.g.*, an office-based lunch break) or location-flexible. Patients sharing the same household may prefer temporally coordinated visits to reduce disruption. Continuity of care is promoted by favoring assignments that preserve the caregiver from previous planning periods. Temporal stability is modeled by penalizing deviations from two reference times: a short-term reference corresponding to the schedule most recently communicated to the patient, and a long-term reference corresponding to the baseline routine established at the start of the care episode.

The planning decisions determine which appointments are served, which caregiver shifts serve them, in what sequence visits and breaks occur, and at what time each service starts. The objective balances operational criteria, including coverage, travel time, waiting time, and overtime, with patient-centered criteria, including caregiver preferences, continuity of care, temporal stability, and household coordination. This integrated representation provides the basis for the computational and managerial analyses developed in Sections 4 and 5.

Table 2 summarizes the notation used in the MILP formulation. All time variables and parameters are expressed as times of day (minutes since midnight) on the planning day. In particular, the time-window bounds $\underline{T}_i, \bar{T}_i$, the service-start variables t_i , and the reference times S_i and O_i share this common representation, so that temporal-stability penalties measure deviations from a patient's daily routine rather than shifts across planning periods.

3.2. MILP formulation

HHC planning is inherently multi-criteria, requiring planners to balance operational efficiency with patient-centered service quality. In practice, candidate schedules are evaluated simultaneously in terms of coverage, travel effort, waiting time, overtime, continuity of care, patient preferences, and schedule predictability. Discussions with our partner organization confirmed that no single metric is sufficient to represent planning quality and that the relative importance of criteria may vary depending on demand conditions and managerial priorities. We therefore adopt a weighted-sum objective function that aggregates multiple performance components². The weights are not intended to encode a single universal preference structure. Rather, they provide a transparent mechanism for exploring how changes in managerial priorities affect coverage, efficiency, continuity, and temporal stability. This formulation provides a flexible decision-support tool and enables explicit trade-off analyses between efficiency and service quality metrics, which we explore in Sections 4 and 5.

Objective function. We minimize a weighted sum of operational and service-quality criteria:

$$\begin{aligned} \min \quad & \alpha_1 \sum_{i \in A} \sigma_i^S (1 - z_i) + \alpha_2 \left(\sum_{k \in K} \sum_{\substack{i, j \in N \\ \{i, j\} \cap B^2 = \emptyset}} \tau_{ijk} x_{ijk} + \sum_{k \in K^2} \sum_{i \in A \cup \{s_k\}} \sum_{j \in A \cup \{e_k\}} \tau_{ijk} \hat{x}_{ijk} \right) \\ & + \alpha_3 \sum_{i \in A} \sum_{k \in K} \zeta_{ik} \sum_{j \in N} x_{ijk} + \alpha_4 \sum_{(i, j) \in A^l} v_{ij} + \alpha_5 \sum_{i \in A} \sum_{k \in K} w_{ik} \end{aligned}$$

²We use a weighted-sum aggregation rather than a lexicographic or ε -constraint formulation because the aim is to quantify trade-offs among operational and patient-centered criteria under a common decision-support framework. Although coverage is the dominant criterion in practice, we do not impose it as a strict lexicographic first priority; Section 5.3 shows that doing so recovers essentially no additional coverage, so the weighted-sum treatment does not sacrifice coverage relative to a coverage-first formulation.

Table 2: Sets, parameters, and decision variables used in the MILP formulation

Sets	
A	Set of appointments ($i \in A$)
A^c	Set of representative appointment pairs (i, j) belonging to the same customer
A^l	Set of representative appointment pairs (i, j) involving co-located patients for which temporal proximity is desired
K	Set of caregiver shifts ($k \in K$)
K_i	Set of shifts corresponding to the previously assigned caregiver(s) for appointment i
R	Set of qualification/requirement types ($r \in R$)
B^1	Set of location-specific break nodes
B^2	Set of location-flexible break nodes
B	Set of all break nodes ($B = B^1 \cup B^2$)
B_k	Set of break nodes required in shift k (possibly empty)
K^1	Set of shifts with a location-specific break
K^2	Set of shifts with a location-flexible break
b_k^1	Location-specific break node of shift k , when present
b_k^2	Location-flexible break node of shift k , when present
S	Set of shift start nodes ($S = \{s_k : k \in K\}$)
E	Set of shift end nodes ($E = \{e_k : k \in K\}$)
N	Set of all nodes ($N = A \cup B \cup S \cup E$)
Parameters	
τ_{ijk}	Travel time from node $i \in N \setminus B^2$ to node $j \in N \setminus B^2$ for shift $k \in K$
σ_i	Service duration of node $i \in N$, with $\sigma_i = 0$ for shift start and end nodes
$[\underline{T}_i, \overline{T}_i]$	Time window within which node $i \in N$ must start service
$[\underline{L}_k, \overline{L}_k]$	Contracted working time window of shift $k \in K$
ρ_{ir}	Equals 1 if appointment $i \in A$ requires qualification $r \in R$, 0 otherwise
q_{kr}	Equals 1 if shift $k \in K$ satisfies qualification $r \in R$, 0 otherwise
ν_i	Number of caregivers required for appointment $i \in A$
ζ_{ik}	Preference or assignment cost of assigning appointment $i \in A$ to shift $k \in K$
S_i	Short-term reference start time of appointment $i \in A$, based on the most recently communicated schedule (time of day)
O_i	Long-term reference start time of appointment $i \in A$, based on the baseline care-episode routine (time of day)
ς	Superlinear exponent ($\varsigma > 1$) on service duration in the unserved-demand penalty
Decision variables	
x_{ijk}	Binary; equals 1 if node $j \in N$ is visited immediately after node $i \in N$ in shift $k \in K$
$\hat{x}_{ijk}$	Binary; equals 1 if node j is visited after node i with the location-flexible break b_k^2 in between, for $k \in K^2$
y_{ij}	Binary; equals 1 if appointment i precedes appointment j , for both directions associated with pairs in A^c
z_i	Binary; equals 1 if appointment $i \in A$ is served
δ_i^{prev}	Binary; equals 1 if appointment $i \in A$ is served by no shift in K_i , and 0 otherwise
t_i	Continuous; start time of service at node $i \in N$ on the planning day
w_{ik}	Continuous; waiting time before starting appointment $i \in A$ in shift $k \in K$
o_k^s	Continuous; early overtime of shift $k \in K$
o_k^e	Continuous; late overtime of shift $k \in K$
Δ_i^t	Continuous; absolute deviation of t_i from the short-term reference time S_i
Δ_i^o	Continuous; absolute deviation of t_i from the long-term reference time O_i
v_{ij}	Continuous; temporal proximity deviation between appointments $(i, j) \in A^l$

$$+ \alpha_6 \sum_{k \in K} (o_k^s + o_k^e) + \alpha_7 \sum_{i \in A} \Delta_i^t + \alpha_8 \sum_{i \in A} \Delta_i^o + \alpha_9 \sum_{i \in A} \delta_i^{\text{prev}}, \quad (1)$$

where $\alpha_1, \dots, \alpha_9$ are nonnegative weights reflecting organizational priorities. The first term penalizes unserved demand. It is weighted by the service duration raised to a mild superlinear exponent ($\varsigma > 1$) rather than by duration alone, so that leaving a single long appointment unserved is penalized more heavily than leaving several short appointments of equivalent total duration; this reflects the provider's practice of prioritizing coverage of longer, typically more care-intensive visits. The second term captures total travel time, separating travel on ordinary route arcs from travel across a location-flexible break. Since a location-flexible break represents an interruption in service time rather than a physical location, arcs incident to b_k^2 do not contribute physical travel time; the travel between the surrounding nodes is counted through $\hat{x}_{ijk}$. The third term penalizes prefer-

ence violations in caregiver–appointment assignments, where ζ_{ik} represents the preference rank of assigning appointment i to shift k (lower values indicating higher preference). The fourth term penalizes temporal gaps between co-located appointments. The fifth and sixth terms penalize waiting time and overtime, respectively. The seventh and eighth terms quantify temporal deviations from reference schedules, distinguishing short-term and long-term stability. The ninth term penalizes continuity-of-care violations through the reassignment indicator δ_i^{prev} , which is zero for unserved appointments.

Routing structure. Each shift $k \in K$ originates at a designated start node s_k and terminates at an end node e_k . If a shift includes mandatory breaks, these must also be incorporated into the route. In addition, each node has at most one outgoing arc within any shift; together with flow conservation, this ensures that no node is visited more than once per route:

$$\sum_{j \in N} x_{s_k j k} = 1, \quad \forall k \in K, \quad (2)$$

$$\sum_{i \in N} x_{i e_k k} = 1, \quad \forall k \in K, \quad (3)$$

$$\sum_{j \in N} x_{b_j k} = 1, \quad \forall k \in K, \quad \forall b \in B_k, \quad (4)$$

$$\sum_{j \in N} x_{i j k} \leq 1, \quad \forall k \in K, \quad \forall i \in N. \quad (5)$$

The start, break, and end nodes are shift-specific. The following constraints ensure that such nodes are used only by their corresponding shift, while also prohibiting arcs leaving end nodes and arcs entering start nodes:

$$\sum_{j \in N} x_{i j k} = 0, \quad \forall k \in K, \quad \forall i \in (\mathcal{S} \cup B \cup E) \setminus (B_k \cup \{s_k\}), \quad (6)$$

$$\sum_{j \in N} x_{j i k} = 0, \quad \forall k \in K, \quad \forall i \in (\mathcal{S} \cup B \cup E) \setminus (B_k \cup \{e_k\}). \quad (7)$$

$$(8)$$

Flow conservation is enforced for all intermediate nodes, namely appointments and breaks belonging to the corresponding shift:

$$\sum_{\substack{j \in N \\ j \neq i}} x_{i j k} - \sum_{\substack{j \in N \\ j \neq i}} x_{j i k} = 0, \quad \forall k \in K, \quad \forall i \in A \cup B_k. \quad (9)$$

Self-loop is prohibited by:

$$\sum_{k \in K} x_{i i k} = 0, \quad \forall k \in K, \quad \forall i \in N. \quad (10)$$

The auxiliary variable $\hat{x}_{ijk}$ indicates whether node j is visited after node i with a location-flexible break b_k^2 in between during shift $k \in K^2$:

$$\hat{x}_{ijk} \geq x_{i b_k^2 k} + x_{b_k^2 j k} - 1, \quad \forall k \in K^2, \quad \forall i \in A \cup \{s_k\}, \quad \forall j \in A \cup \{e_k\}, \quad (11)$$

$$\hat{x}_{ijk} \leq x_{i b_k^2 k}, \quad \forall k \in K^2, \quad \forall i \in A \cup \{s_k\}, \quad \forall j \in A \cup \{e_k\}, \quad (12)$$

$$\hat{x}_{ijk} \leq x_{b_k^2 j k}, \quad \forall k \in K^2, \quad \forall i \in A \cup \{s_k\}, \quad \forall j \in A \cup \{e_k\}. \quad (13)$$

Appointment coverage and synchronization. If appointment i is served ($z_i = 1$), it must be assigned to exactly ν_i shifts. Thus, standard appointments are covered by one caregiver, whereas synchronized appointments requiring multiple caregivers are assigned to the required number of shifts. Since all shifts visiting appointment i share the same service start time variable t_i , simultaneous service is enforced by construction:

$$\sum_{k \in K} \sum_{\substack{j \in N \\ j \neq i}} x_{ijk} = \nu_i z_i, \quad \forall i \in A. \quad (14)$$

Qualification feasibility. A shift can serve an appointment only if it satisfies all required qualifications. Thus, whenever appointment i requires qualification r , assignment to a shift lacking that qualification is forbidden:

$$\sum_{j \in N} x_{ijk} \leq q_{kr} + (1 - \rho_{ir}), \quad \forall i \in A, \forall k \in K, \forall r \in R. \quad (15)$$

Time feasibility. Service start times must be consistent with service durations and travel times along each route. For direct arcs that do not involve a location-flexible break, we impose:

$$t_i + \sigma_i + \tau_{ijk} \leq t_j + M_{ijk}^T(1 - x_{ijk}), \quad \forall k \in K, \forall i \in A \cup B_k \cup \{s_k\}, \forall j \in A \cup B_k \cup \{e_k\} : \{i, j\} \cap B^2 = \emptyset. \quad (16)$$

When a location-flexible break b_k^2 is inserted between two nodes, the break represents an interruption in service time rather than a physical location. The following constraints place the break between its surrounding nodes and ensure that the elapsed time includes both direct travel between these nodes and the break duration:

$$t_i + \sigma_i \leq t_{b_k^2} + M_{ib_k^2}^\sigma(1 - x_{ib_k^2}), \quad \forall k \in K^2, \forall i \in A \cup B_k \cup \{s_k\}, \quad (17)$$

$$t_{b_k^2} + \sigma_{b_k^2} \leq t_j + M_{b_k^2 j}^\sigma(1 - x_{b_k^2 j}), \quad \forall k \in K^2, \forall i \in A \cup B_k \cup \{e_k\}, \quad (18)$$

$$t_i + \sigma_i + \sigma_{b_k^2} + \tau_{ijk} \leq t_j + M_{ijk}^{\tau_b} \left(2 - x_{ib_k^2} - x_{b_k^2 j} \right), \quad (19)$$

$$\forall k \in K^2, \forall i \in A \cup \{s_k\}, \forall j \in A \cup \{e_k\}.$$

Time windows must be respected at all nodes:

$$\underline{T}_i \leq t_i \leq \bar{T}_i, \quad \forall i \in N. \quad (20)$$

The big- M constants are defined as $M_{ijk}^T = \max \{0, \bar{T}_i + \sigma_i + \tau_{ijk} - \underline{T}_j\}$, $M_{ijk}^{\tau_b} = \max \{0, M_{ijk}^T + \sigma_{b_k^2}\}$, and $M_{ij}^\sigma = \max \{0, \bar{T}_i + \sigma_i - \underline{T}_j\}$.

Same customer. Appointments belonging to the same customer must not overlap in time. For each pair $(i, j) \in A^c$, the binary variables y_{ij} and y_{ji} determine their relative ordering. If both appointments are served, exactly one of the two possible orderings must hold; otherwise both ordering variables are forced to zero:

$$y_{ij} + y_{ji} \geq z_i + z_j - 1, \quad \forall (i, j) \in A^c, \quad (21)$$

$$y_{ij} + y_{ji} \leq z_i, \quad \forall (i, j) \in A^c, \quad (22)$$

$$y_{ij} + y_{ji} \leq z_j, \quad \forall (i, j) \in A^c. \quad (23)$$

Then, if $y_{ij} = 1$, appointment i must be completed before appointment j begins:

$$t_i + \sigma_i \leq t_j + M_{ij}^\sigma(1 - y_{ij}), \quad \forall (i, j) \in A^c, \quad (24)$$

$$t_j + \sigma_j \leq t_i + M_{ji}^\sigma(1 - y_{ji}), \quad \forall (i, j) \in A^c, \quad (25)$$

Household temporal coordination. Patients sharing the same household may prefer visits to occur close in time. For each pair of co-located appointments $(i, j) \in A^l$, the variable v_{ij} measures the temporal gap between their service intervals. The gap is zero when the intervals overlap and positive when one visit finishes before the other begins. The constraints are activated only when both appointments are served:

$$v_{ij} \geq t_i - (t_j + \sigma_j) - M_{ij}^v(2 - z_i - z_j), \quad \forall (i, j) \in A^l, \quad (26)$$

$$v_{ij} \geq t_j - (t_i + \sigma_i) - M_{ji}^v(2 - z_i - z_j), \quad \forall (i, j) \in A^l. \quad (27)$$

where $M_{ij}^v = \max \{0, \bar{T}_i - (\underline{T}_j + \sigma_j)\}$.

Waiting time. Waiting time before serving appointment i in shift k captures the idle time between a caregiver's arrival at the appointment location and the start of service. Three cases arise depending on the arc structure. For two directly connected nodes:

$$w_{ik} \geq t_i - (t_j + \sigma_j + \tau_{jik}) - M_{jik}^w(1 - x_{jik}), \quad \forall k \in K, \forall i \in A, \forall j \in A \cup \{s_k\}. \quad (28)$$

When a location-flexible break is inserted between nodes j and i :

$$w_{ik} \geq t_i - (t_j + \sigma_j + \tau_{jik}) - M_{jik}^w(2 - x_{jb_k^2k} - x_{b_k^2ik}), \quad \forall k \in K^2, \forall i \in A, \forall j \in A \cup \{s_k\}. \quad (29)$$

When a location-specific break is inserted between nodes j and i , the travel follows a compound path through the break location:

$$w_{ik} \geq t_i - (t_j + \sigma_j + \tau_{jb_k^1k} + \tau_{b_k^1ik}) - M_{jik}^{wb}(2 - x_{jb_k^1k} - x_{b_k^1ik}), \quad \forall k \in K^1, \forall i \in A, \forall j \in A \cup \{s_k\}. \quad (30)$$

where $M_{jik}^w = \max \{0, \bar{T}_i - (\underline{T}_j + \sigma_j + \tau_{jik})\}$ and $M_{jik}^{wb} = \max \{0, \bar{T}_i - (\underline{T}_j + \sigma_j + \tau_{jb_k^1k} + \tau_{b_k^1ik})\}$. In the break cases in particular, penalizing total waiting discourages idle time from arising on both sides of an inserted break, favouring schedules that concentrate any unavoidable slack on one side of the break where the time windows allow. The model aggregates waiting time across the route, as the operational cost to the HHC provider depends on total caregiver idle time rather than on where this idle time occurs.

Overtime. Overtime is computed relative to the contracted shift window, quantifying time incurred by starting before $\underline{L}_k$ or ending after $\bar{L}_k$:

$$o_k^s \geq \underline{L}_k - t_{s_k}, \quad o_k^e \geq t_{e_k} - \bar{L}_k, \quad \forall k \in K. \quad (31)$$

Temporal stability. Short-term and long-term temporal deviations are modeled using absolute-value linearizations. For served appointments ($z_i = 1$), Δ_i^t captures the absolute deviation from the most recently communicated appointment time S_i , while Δ_i^o captures the deviation from the baseline routine time O_i . For unserved appointments ($z_i = 0$), the big- M terms deactivate the constraints:

$$\Delta_i^t \geq t_i - S_i - M_i^\Delta(1 - z_i), \quad \Delta_i^t \geq S_i - t_i - M_i^\Delta(1 - z_i), \quad \forall i \in A, \quad (32)$$

$$\Delta_i^o \geq t_i - O_i - M_i^\Delta(1 - z_i), \quad \Delta_i^o \geq O_i - t_i - M_i^\Delta(1 - z_i), \quad \forall i \in A. \quad (33)$$

where $M_i^\Delta = \max \{0, \bar{T}_i - \underline{T}_i\}$.

Continuity of care. The reassignment indicator δ_i^{prev} is linked to whether appointment i is served by one of its previously assigned caregivers (the shifts in K_i):

$$\sum_{j \in N} \sum_{k \in K_i} x_{ijk} \geq z_i - \delta_i^{\text{prev}}, \quad \forall i \in A, \quad (34)$$

If appointment i is served ($z_i = 1$), then $\delta_i^{\text{prev}} = 0$ is possible only if it is assigned to a previously assigned caregiver. Otherwise, $\delta_i^{\text{prev}} = 1$ and a reassignment penalty is incurred.

Strengthening inequalities. To improve computational performance, we add strengthening constraints that do not alter the feasible set.

We define the set $\Omega \subseteq A \times K$ of provably infeasible appointment-shift pairs through a preprocessing procedure that checks three conditions:

- *Qualification infeasibility:* pair (i, k) is infeasible if shift k lacks a required qualification of i , i.e., there exists $r \in R$ with $\rho_{ir} = 1$ and $q_{kr} = 0$.
- *Travel-aware temporal infeasibility:* pair (i, k) is infeasible if appointment i cannot be started within its time window while allowing shift k to travel from its start node and return to its end node. A sufficient preprocessing test is:

$$\max\{\underline{T}_i, \underline{T}_{s_k} + \tau_{s_k i k}\} > \min\{\bar{T}_i, \bar{T}_{e_k} - \sigma_i - \tau_{i e_k k}\}.$$

where $\underline{T}_{s_k}$ and $\bar{T}_{e_k}$ denote the earliest feasible start and latest feasible end of shift k . These bounds are set to the widest span the shift may use, i.e. the contracted window $[\underline{L}_k, \bar{L}_k]$ extended by the permitted early-start and late-end overtime, so that the test admits any appointment that shift k could serve once overtime is used and excludes only pairs that remain infeasible even at maximum usable capacity.

- *Break conflict infeasibility:* if shift k has a mandatory break $b_k \in [b_k^1, b_k^2]$ with time window $[\underline{T}_{b_k}, \bar{T}_{b_k}]$ and duration σ_{b_k} , the break may split the available service time into pre-break and post-break feasible regions. If appointment i cannot be feasibly scheduled in either region, the pair (i, k) is added to Ω .

We exclude all such assignments:

$$\sum_{j \in N} x_{ijk} = 0, \quad \forall (i, k) \in \Omega. \quad (35)$$

Moreover, if for some appointment $i \in A$ every shift $k \in K$ yields $(i, k) \in \Omega$, then i is declared unplannable and removed from the active appointment set before optimization, with its service duration added to the unserved-demand component of the objective.

Domains.

$$x_{ijk} \in \{0, 1\}, \quad \forall i, j \in N, \forall k \in K, \quad (36)$$

$$\hat{x}_{ijk} \in \{0, 1\}, \quad \forall k \in K^2, \forall i \in A \cup \{s_k\}, \forall j \in A \cup \{e_k\}, \quad (37)$$

$$y_{ij}, y_{ji} \in \{0, 1\}, \quad \forall (i, j) \in A^c, \quad (38)$$

$$z_i, \delta_i^{\text{prev}} \in \{0, 1\}, \quad \forall i \in A, \quad (39)$$

$$t_i \geq 0, \quad \forall i \in N, \quad (40)$$

$$w_{ik} \geq 0, \quad \forall i \in A, \forall k \in K, \quad (41)$$

$$o_k^s, o_k^e \geq 0, \quad \forall k \in K, \quad (42)$$

$$\Delta_i^t, \Delta_i^o \geq 0, \quad \forall i \in A, \quad (43)$$

$$v_{ij} \geq 0, \quad \forall (i, j) \in A^l. \quad (44)$$

3.3. Hexaly formulation

To complement the MILP of Section 3.2, we present a solver-native formulation in Hexaly Optimizer (Benoist et al., 2011; Gardi et al., 2014). This formulation expresses routing decisions through per-shift list variables and timing decisions through interval variables. Its purpose is to provide a transparent and reproducible commercial-solver benchmark for the same planning problem, rather than to introduce a separate algorithmic contribution. The computational performance of the MILP and Hexaly formulations is compared in Section 4.

The Hexaly formulation retains the sets and parameter notation of Section 3.1. For each shift $k \in K$, the main routing decision is a list variable $\mathcal{L}_k$, which contains the nodes assigned to shift k in visit order. Its length $|\mathcal{L}_k|$ is itself a decision and may differ across shifts. Whether a node i belongs to shift k is recovered by $\text{contains}(\mathcal{L}_k, i) \in \{0, 1\}$, and its position by $\text{index}(\mathcal{L}_k, i) \in \{0, \dots, |\mathcal{L}_k| - 1\}$. For each node $i \in N$, an interval variable $\mathcal{I}_i$ is defined over the start-time domain $[\underline{T}_i, \overline{T}_i]$.

For appointments requiring multiple caregivers, the implementation represents appointment i through ν_i route copies while retaining a single interval variable $\mathcal{I}_i$ for the original appointment. This ensures that all caregivers assigned to a synchronized appointment share the same service start and end times. For notational simplicity, the expressions below are written in terms of the original appointment index i ; in the implementation, membership expressions for synchronized appointments are evaluated over the corresponding route copies. In the notation below, nv_{ik} denotes whether appointment i is assigned to shift k , and $\text{nv}_i = \sum_{k \in K} \text{nv}_{ik}$ denotes the total number of assigned caregivers:

$$\text{nv}_i = \sum_{k \in K} \text{nv}_{ik}, \quad z_i = 1[\text{nv}_i > 0]. \quad (45)$$

We adopt the conventional notation $a ? b : c$ for the conditional expression returning b when a is true and c otherwise.

Objective function. The objective components are aggregated identically to the MILP, yielding the weighted-sum objective

$$\begin{aligned} \min \quad & \alpha_1 \sum_{i \in A} (1 - z_i) \sigma_i^s + \alpha_2 \sum_{k \in K} \text{travel}_k + \alpha_3 \sum_{i \in A} \sum_{k \in K} \zeta_{ik} \text{nv}_{ik} + \alpha_4 \sum_{(i,j) \in A^l} v_{ij} \\ & + \alpha_5 \sum_{i \in A} \sum_{k \in K} w_{ik} + \alpha_6 \sum_{k \in K} (o_k^s + o_k^e) + \alpha_7 \sum_{i \in A} \Delta_i^t + \alpha_8 \sum_{i \in A} \Delta_i^o + \alpha_9 \sum_{i \in A} \delta_i^{\text{prev}}. \end{aligned} \quad (46)$$

This objective corresponds term-by-term to Equation (1). The derived quantities travel_k , w_{ik} , o_k^s , o_k^e , v_{ij} , Δ_i^t , Δ_i^o , and δ_i^{prev} are defined below.

Routing structure. A single global constraint ensures that the lists $\mathcal{L}_1, \dots, \mathcal{L}_{|K|}$ form a partition of the visited nodes:

$$\text{disjoint}(\mathcal{L}_1, \dots, \mathcal{L}_{|K|}). \quad (47)$$

Hexaly enforces this constraint natively.

The disjoint Constraint (47) automatically prevents start, end, and break nodes of one shift from appearing in another, so the analog of Constraints (6)-(7) is implicit. Shift-specific start, break, and end nodes are pinned to their list positions (Constraints (2)-(4)):

$$\text{contains}(\mathcal{L}_k, s_k) = 1, \quad \text{index}(\mathcal{L}_k, s_k) = 0, \quad \forall k \in K, \quad (48)$$

$$\text{contains}(\mathcal{L}_k, e_k) = 1, \quad \text{index}(\mathcal{L}_k, e_k) = |\mathcal{L}_k| - 1, \quad \forall k \in K, \quad (49)$$

$$\text{contains}(\mathcal{L}_k, b) = 1, \quad 1 \leq \text{index}(\mathcal{L}_k, b) \leq |\mathcal{L}_k| - 2, \quad \forall k \in K, \forall b \in B_k. \quad (50)$$

Flow conservation (9) and forbidden arcs (5) are guaranteed by the list semantics; each node in $\mathcal{L}_k$ has exactly one predecessor and one successor, except the first and last.

Appointment coverage and synchronization. A node i is served if and only if it appears in some list. Standard appointments must appear in exactly one list, while synchronized appointments that require ν_i caregivers must appear in exactly ν_i lists or in none (Constraints (14)):

$$n\nu_{ik} \leq 1 \quad \forall i \in A, k \in K, \quad (51)$$

$$n\nu_i \in \{0, \nu_i\}, \quad \forall i \in A. \quad (52)$$

Since the interval variable $\mathcal{I}_i$ is defined once per appointment rather than once per copy, all caregivers assigned to a synchronized appointment share the same start and end times by construction.

Qualification feasibility. Qualification feasibility (Constraints (15)) is expressed by forbidding i from any shift that lacks a required qualification:

$$n\nu_{ik} = 0, \quad \forall (i, k) \in \{(i, k) \in A \times K \mid \exists r \in R : \rho_{ir} = 1, q_{kr} = 0\}. \quad (53)$$

Time feasibility. Service intervals are constrained to have the right length:

$$\text{end}(\mathcal{I}_i) - \text{start}(\mathcal{I}_i) = \sigma_i, \quad \forall i \in N. \quad (54)$$

Constraints (16)-(19), which use big- M coefficients to enforce travel time between consecutive nodes, are replaced by a quantified constraint over each list. Let $I[i]$ denote the interval indexed by node i , so that $I[\mathcal{L}_k[p]]$ is the interval of the node at position p of $\mathcal{L}_k$.

$$\text{end}(I[\mathcal{L}_k[p]]) + \tau_{\mathcal{L}_k[p], \mathcal{L}_k[p+1], k} \leq \text{start}(I[\mathcal{L}_k[p+1]]), \quad \forall k \in K, \forall p \in [0, |\mathcal{L}_k| - 2] : \mathcal{L}_k[p], \mathcal{L}_k[p+1] \neq b_k^2. \quad (55)$$

Constraints (55) are over the dynamic length of $\mathcal{L}_k$; it is evaluated only at the consecutive pairs that actually occur in the route. When a location-flexible break is inserted, the segment immediately following the break uses the direct travel time between the surrounding appointments rather than a compound path through the break location:

$$\text{end}(I[i]) + \sigma_{b_k^2} + \tau_{i, j, k} \leq \text{start}(I[j]), \quad \text{with } i = \mathcal{L}_k[\text{index}(\mathcal{L}_k, b_k^2) - 1], j = \mathcal{L}_k[\text{index}(\mathcal{L}_k, b_k^2) + 1]. \quad (56)$$

In addition, time windows (20) are imposed directly through the domain of each interval variable.

Same customer. The same-customer ordering constraints (24)-(25) are replaced by an interval-level disjunction:

$$\neg(z_i \wedge z_j) \vee (\text{end}(\mathcal{I}_i) \leq \text{start}(\mathcal{I}_j) \vee \text{end}(\mathcal{I}_j) \leq \text{start}(\mathcal{I}_i)), \quad \forall (i, j) \in A^c. \quad (57)$$

Household temporal coordination. The household proximity expression v_{ij} for $(i, j) \in A^l$ (Constraints (26)-(27)) is computed directly from the intervals when both appointments are served:

$$v_{ij} = (z_i \wedge z_j) ? \max(0, \text{start}(\mathcal{I}_i) - \text{end}(\mathcal{I}_j), \text{start}(\mathcal{I}_j) - \text{end}(\mathcal{I}_i)) : 0. \quad (58)$$

Per-shift travel time. The total travel time of shift k is calculated as:

$$\text{travel}_k = \sum_{p=1}^{|\mathcal{L}_k|-1} \begin{cases} \tau_{\mathcal{L}_k[p-1], \mathcal{L}_k[p+1], k} & \text{if } \mathcal{L}_k[p] = b_k^2, \\ 0 & \text{if } \mathcal{L}_k[p-1] = b_k^2, \\ \tau_{\mathcal{L}_k[p-1], \mathcal{L}_k[p], k} & \text{otherwise.} \end{cases} \quad \forall k \in K. \quad (59)$$

Indexing the travel-time array by the decision expression $\mathcal{L}_k[p]$ is supported natively. When the location-flexible break b_k^2 occupies a position in $\mathcal{L}_k$, the step entering the break is charged the direct travel between the node preceding and the node following the break, and the step leaving the break is charged nothing, so the break crossing contributes a single physical travel time between its neighbours; all other steps are charged the ordinary travel time.

Waiting time. Waiting time at position p of $\mathcal{L}_k$ (Constraints (28)-(30)) is:

$$w_{\mathcal{L}_k[p], k} = \max(0, \text{start}(I[\mathcal{L}_k[p]]) - \text{end}(I[\mathcal{L}_k[p-1]]) - \tau_{\mathcal{L}_k[p-1], \mathcal{L}_k[p], k}), \quad \forall k \in K, \forall p \in [1, |\mathcal{L}_k| - 2] : \mathcal{L}_k[p], \mathcal{L}_k[p-1] \notin B_k. \quad (60)$$

Here waiting is the idle time between the caregiver's arrival at $\mathcal{L}_k[p]$, determined by the predecessor's completion and the direct travel, and the start of service. This case applies whenever neither the current nor the preceding position is a break.

$$w_{\mathcal{L}_k[p], k} = \max(0, \text{start}(I[\mathcal{L}_k[p]]) - \text{end}(I[\mathcal{L}_k[p-2]]) - \tau_{\mathcal{L}_k[p-2], \mathcal{L}_k[p], k}), \quad \forall k \in K^2 : \mathcal{L}_k[p-1] = b_k^2. \quad (61)$$

When position p immediately follows the location-flexible break b_k^2 , the break carries no location, so the predecessor is taken as the node before the break, $\mathcal{L}_k[p-2]$ and the direct travel from that node to the following is charged. The break duration is counted as waiting following the logic implemented in MILP.

$$w_{\mathcal{L}_k[p], k} = \max(0, \text{start}(I[\mathcal{L}_k[p]]) - \text{end}(I[\mathcal{L}_k[p-2]]) - \tau_{\mathcal{L}_k[p-2], b_k^1, k} - \tau_{b_k^1, \mathcal{L}_k[p], k}), \quad \forall k \in K^1 : \mathcal{L}_k[p-1] = b_k^1. \quad (62)$$

When position p follows the location-specific break b_k^1 , the caregiver's arrival is computed along the compound path $\mathcal{L}_k[p-2] \rightarrow b_k^1 \rightarrow \mathcal{L}_k[p]$ through the break location. As in the flexible case, the break duration itself is included in waiting.

In all three cases waiting is evaluated only at appointment positions, and the three cases reproduce the MILP waiting constraints (28)-(30).

Overtime. Overtime follows by (Constraints (31)):

$$o_k^s = \max(0, \underline{L}_k - \text{start}(\mathcal{I}_{s_k})), \quad o_k^e = \max(0, \text{end}(\mathcal{I}_{e_k}) - \bar{L}_k), \quad (63)$$

Temporal stability. Temporal stability (Constraints (32)-(33)) is encoded as conditional absolute deviations:

$$\Delta_i^t = z_i ? |\text{start}(\mathcal{I}_i) - S_i| : 0, \quad \forall i \in A, \quad (64)$$

$$\Delta_i^o = z_i ? |\text{start}(\mathcal{I}_i) - O_i| : 0, \quad \forall i \in A. \quad (65)$$

Continuity of care. Continuity-of-care reassignment is recovered from list membership:

$$\delta_i^{\text{prev}} = z_i ? 1 \left[\sum_{k' \in K_i} \text{nv}_{i,k'} = 0 \right] : 0, \quad \forall i \in A. \quad (66)$$

The indicator equals 1 when appointment i is served but by no previously assigned caregiver, and 0 when appointment i is either unserved or assigned to at least one caregiver in K_i .

4. Computational study

This section evaluates the proposed formulations on operational instances and addresses three questions. First, do the MILP and Hexaly formulations produce consistent results on instances for which the optimum is known? Second, can they generate strong feasible solutions at the operational scale within a time budget compatible with daily planning? Third, when the optimum is unavailable, how much confidence can be placed in the quality of the resulting incumbents?

To address the third question, we use a genetic algorithm (GA) and an adaptive large neighborhood search (ALNS) matheuristic as independent benchmarks. They broaden the search for good feasible solutions and provide an empirical check on the quality of the MILP and Hexaly incumbents. For each instance, the best objective value obtained across the MILP, Hexaly, GA, and ALNS runs and all random seeds is retained as the best-known feasible value. The distance to this reference indicates how close each incumbent is to the strongest solution found, but should not be interpreted as an optimality gap because the reference is not proven optimal. For the stochastic approaches, median results across random seeds are also reported to characterize the performance of a typical run. Further implementation details are provided in [Appendix A](#).

The remainder of the section is organized as follows. Sections [4.1](#) and [4.2](#) describe the data and experimental design. Section [4.3](#) reports the small-instance verification, Section [4.4](#) evaluates the preprocessing-based variable fixing, and Sections [4.5](#) and [4.6](#) examine the operational time budget and comparative performance.

4.1. Data description

We evaluate the proposed formulations on 31 daily instances from January 2025. The operational data were supplied by an HHC organization that uses the planning software developed by our Swiss industry partner. Each instance specifies the appointment set A and the shift set K . For each shift, the data include the contracted working window $[\underline{L}_k, \bar{L}_k]$ and shift-specific start and end nodes. The instances also include the qualification parameters (ρ_{ir}, q_{kr}) , the reference times S_i and O_i , and a precomputed table of pairwise travel durations indexed by origin, destination, and transportation mode. The shift-dependent travel times τ_{ijk} are obtained from this table according to the transportation mode associated with shift k .

The long-term reference O_i is taken from the baseline start time recorded at the beginning of the corresponding care episode. For each recurring appointment, the short-term reference S_i is reconstructed from the HHC partner's historical schedules and set to the most recently scheduled start time in the same appointment series. The previous-caregiver set K_i is identified from the same

historical records. These quantities are fixed before each daily instance is solved. Each operating day is therefore treated as a static planning problem defined relative to the operational history available at that time.

Two small verification instances, *Small test 1* and *Small test 2*, are used in Section 4.3. Both can be solved to proven optimality by the MILP within seconds, providing known optimal objective values against which the other implementations can be compared.

Table 3 summarizes the main instance characteristics. The complete daily breakdown is reported in Table A.1 in Appendix B. The instances vary considerably in scale over the planning month. The number of appointments $|A|$ ranges from 103 to 219, while the number of shifts $|K|$ ranges from 8 to 18. The instances include up to 154 distinct customers and 37 qualification types.

We define the demand-to-capacity ratio as the total requested service time divided by the total contracted caregiver time. Across the 31 days, this ratio ranges from 53% to 134%, with a median of 97%. This shows that service demand is already close to the available contracted capacity on many days.

This ratio does not include travel time or account for qualification compatibility, break requirements, or time windows. Adding the travel time required to serve visits to the service demand (estimated by applying the average travel time per appointment to other requested appointments) the median combined service-and-travel workload corresponds to 112.6% of contracted capacity. This indicates that the instances are highly constrained even before the full scheduling problem is considered.

Table 3: Summary of instance characteristics across the 31 operational instances. Minimum, median, and maximum values are reported. The full per-instance breakdown is reported in Table A.1 of Appendix B.

Characteristic	Min	Median	Max
Appointments $ A $	103	176	219
Shifts $ K ^\dagger$	8	13	18
Customers	69	114	154
Qualification types $ R $	15	33	37
Demand-to-capacity (%)	53	97	134
Previous-caregiver share (%)	0.0	24.9	71.4
Synchronized share (%) †	0.5	0.6	1.0
Household pairs	0	2	3
Time-window width, median (min)	80	90	100
Travel time, median (min)	3.6	4.3	4.8

† Reported for the feature-complete instances used in the computational study, *i.e.* after the two controlled modifications.

Most features considered in the model are present directly in the operational data. Depending on the day, between 0% and 71% of appointments have a previous-caregiver reference. The data also contain up to three pairs of appointments located in the same household. For recurring appointments, the baseline and short-term start-time references are available from the historical schedules. These data allow the continuity, household-coordination, and temporal-stability components of the model to be evaluated.

Two features are not represented directly in the January 2025 records. First, lunch breaks are recorded indirectly by splitting each caregiver’s working day into two half-day shifts. Their timing is therefore fixed in the original data. Second, the records contain no appointments requiring two caregivers to arrive simultaneously.

To include these two features in the computational experiments, we make two controlled mod-

ifications. Each pair of half-day shifts belonging to the same caregiver is combined into one shift with a schedulable break. This changes the representation of the working day but preserves the caregiver’s total working time. We also designate one recurring appointment series with a three-hour time window as requiring two simultaneous caregivers. This second modification increases the staffing requirement for that appointment series but leaves its duration, location, and time window unchanged. All other appointment and caregiver data remain unchanged.

4.2. Experimental design

The computational study compares two MILP variants, the list- and interval-based formulation solved with Hexaly, and two heuristic benchmarks. Both MILP variants are solved with Gurobi. The strengthened variant includes the preprocessing-based variable fixing described in Section 4.4 while the Baseline MILP omits this fixing. Comparing the two variants shows whether the preprocessing improves the feasible solutions found, the solver bound, or both. A GA and an ALNS matheuristic provide independent searches for good feasible solutions.

The 31 operational instances are divided chronologically into training and evaluation sets. The first 10 days, January 1–10, 2025, are used to configure the GA and ALNS parameters. The remaining 21 days, January 11–31, form the evaluation set on which the methods are compared. This split prevents the evaluation instances from influencing the parameter tuning and preserves the chronological order of the data.

All methods use the weighted objective defined in Equation (1). The coefficients used in the operational experiments are reported in Table 4. The MILP and Hexaly formulations optimize this objective directly, while the GA and ALNS use it to evaluate candidate solutions.

Table 4: Objective coefficients used in the operational experiments. The coefficients correspond to the operational configuration supplied by our Swiss industry partner and give the greatest priority to coverage. The coverage exponent is set to $\varsigma = 1.1$. Sensitivity to these coefficients is examined in Sections 5.1–5.3

Weight	Objective component	Value	Component unit
α_1	Unserved demand	250	min^ς
α_2	Travel time	10	min
α_3	Assignment preference violation	50	rank units
α_4	Household coordination	0.2	min
α_5	Waiting time	10	min
α_6	Overtime	100	min
α_7	Short-term deviation Δ^t	0.5	min
α_8	Long-term deviation Δ^o	0.1	min
α_9	Caregiver reassignment	10	per visit

The short-term reference S_i and the previous-caregiver set K_i are derived from previously communicated schedules. If each method generated these references from its own earlier solutions, the instances solved on later days would differ across methods. We therefore construct the references once and use the same values for all methods. This ensures that every method solves the same daily instance and that the per-day comparisons remain paired. The long-term references O_i are fixed and common to all methods.³

³As a robustness check, the references were also regenerated separately from each method’s previous solutions. This did not materially change the comparative results.

The GA and ALNS have algorithm-specific control parameters. These parameters are configured on the 10 training instances using iterated racing (López-Ibáñez et al., 2016), with a budget of 1,000 algorithm evaluations and a maximum wall-clock time of 600 seconds per evaluation. Hexaly is run with its default parameter settings. Table 5 reports the search ranges and selected configurations.

Table 5: Search ranges and configurations selected by **irace** on the 10 training instances.

Method	Parameter	Search range	Selected
GA	Population size	[50, 400]	86
	Crossover probability p_c	[0.60, 1.00]	0.66
	Timing mutation p_m^t	[0.05, 0.50]	0.293
	Assignment mutation p_m^a	[0.05, 0.50]	0.139
	SBX index η_c	[5, 30]	16.9
	Mutation index η_m	[5, 30]	25.8
	Tournament size	[2, 5]	3
	Elite fraction	[0.02, 0.20]	0.169
ALNS	Minimum destroy fraction $q_{\min}^{\text{frac}}$	[0.03, 0.15]	0.033
	Maximum destroy fraction $q_{\max}^{\text{frac}}$	[0.10, 0.40]	0.106
	Reward σ_1	[10, 50]	49.7
	Reward σ_2	[1, 20]	9.96
	Reward σ_3	[1, 20]	11.2
	Reaction factor ρ_w	[0.05, 0.50]	0.420
	Segment length L_{seg}	[20, 100]	44
	SA temperature factor κ	[0.001, 0.02]	0.0091
	SA cooling factor ρ_T	[0.990, 0.9995]	0.993

The two MILP variants are deterministic and are each executed once per instance. Hexaly, the GA, and the ALNS contain randomized search components and are each executed under 30 independent random seeds per instance. For these methods, the median objective value describes the performance of a typical run. The best objective value across seeds is reported separately and is used to construct the best-known feasible reference.

Because the optimum is unavailable for the operational instances, we construct a best-known feasible value for each instance. Let f^{BKS} denote the lowest objective value obtained across both MILP variants and all Hexaly, GA, and ALNS runs. For a feasible solution with objective value f , its relative deviation from this reference is calculated as

$$\frac{f - f^{\text{BKS}}}{f^{\text{BKS}}}.$$

This quantity measures the distance to the strongest feasible solution found. It is not an optimality gap because f^{BKS} is not proven optimal.

In the repeated-search comparison, each stochastic method is represented by its best objective value across the 30 seeds, while each deterministic MILP variant is represented by its single run. Median results are reported separately to describe typical stochastic performance. Statistical differences across the evaluation instances are assessed using the Friedman test. When the null hypothesis of equal performance is rejected, pairwise Wilcoxon signed-rank tests are applied to the paired per-instance results, with Holm correction at the 0.05 level. The statistical comparison therefore concerns repeated-search performance rather than the outcome of a single stochastic run.

All experiments are performed on a compute node equipped with dual AMD EPYC 7402 processors. The MILP is solved with Gurobi 11.0.3 and the list- and interval-based formulation with Hexaly 14.5. Each run uses a single thread to ensure comparable runtime measurements.

4.3. Verification on small instances

Before moving to the operational instances, we check the implementations on two small instances that the MILP solves to proven optimality in under one second. The resulting optimal objective values provide a reference for evaluating the Hexaly formulation and the two heuristic benchmarks. Hexaly, the GA, and the ALNS are each run under 30 independent random seeds. All resulting solutions are scored using the objective in Equation (1). Table 6 reports the best and median gap to the MILP optimum, together with the coefficient of variation across seeds.

Table 6: Results on two small instances solved to proven optimality by the MILP. For each randomized method, the table reports the best and median gap to the optimum and the coefficient of variation across 30 seeds.

Instance	Method	Best gap (%)	Median gap (%)	CV (%)
Small test 1	GA	+0.03	+0.27	0.13
	ALNS	+0.03	+0.03	0.00
	Hexaly	+0.05	+0.05	1.11
Small test 2	GA	+0.59	+1.36	0.40
	ALNS	+1.20	+1.20	4.41
	Hexaly	+0.00	+407.8	60.57

The GA and ALNS return solutions close to the optimum on both instances. The best gaps do not exceed 1.2%, and the median gaps do not exceed 1.36%. These results provide a useful check of the objective evaluation and feasibility-handling procedures used by the heuristic implementations.

Hexaly also performs consistently on *Small test 1*. Its behavior on *Small test 2* is more sensitive to the random seed. Seven runs attain the optimum, seven converge to an objective value about 38.1% above the optimum, and the remaining 16 have a gap of approximately 407.8%. The highest-objective runs leave one appointment unserved, which incurs a large coverage penalty. These results show why both median and best-of-seed performance are reported in the operational experiments.

Figure 1 shows the complete distributions across seeds. The three methods are concentrated near the optimum on *Small test 1*. On *Small test 2*, the GA remains tightly concentrated and the ALNS stays relatively close to the optimum, while the Hexaly results fall into three distinct groups. Overall, the small-instance experiments provide a consistency check on the implementations and support the repeated-seed protocol used for the randomized methods.

4.4. Formulation strengthening

We evaluate the preprocessing-based variable fixing introduced in Constraint (35). For each appointment-shift pair $(i, k) \in \Omega$ identified as infeasible, the corresponding assignment arcs are fixed to zero. A pair is excluded because of qualification incompatibility, travel-aware temporal infeasibility, or incompatibility with the break constraints. Since only assignments that cannot belong to a feasible schedule are removed, the preprocessing does not exclude any feasible schedule or change the optimal objective value.

We compare the strengthened and baseline MILP formulations on all 31 operational instances using a time limit of 1,800 seconds. The strengthened formulation includes Constraint (35), while the Baseline MILP omits it. On average, the number of nonzeros in the presolved model decreases

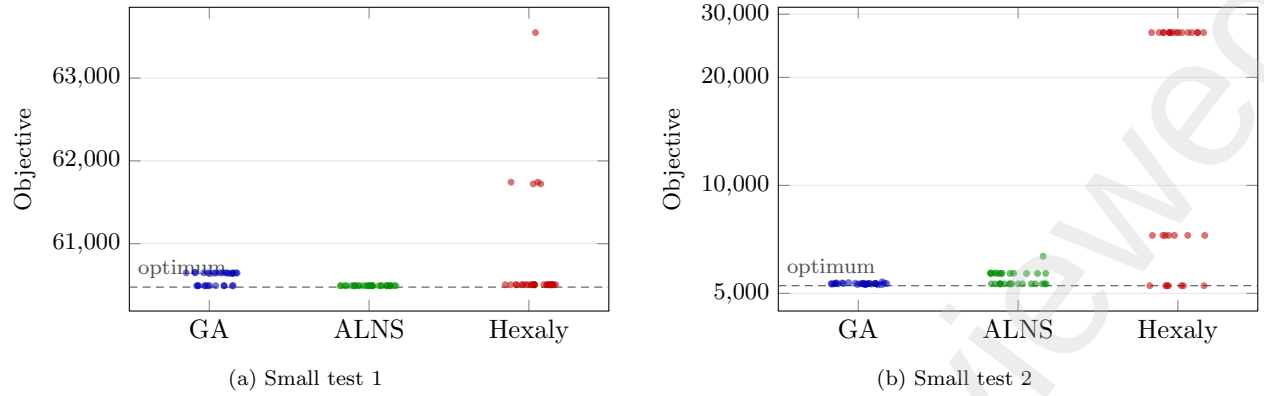


Figure 1: Objective values over 30 random seeds for each randomized method on the two verification instances. Each point represents one run and the dashed line indicates the MILP optimum. The vertical axis is linear for *Small test 1* and logarithmic for *Small test 2*. The three distinct Hexaly objective levels on *Small test 2* show its sensitivity to the random seed on this instance.

from 3.39 million to 1.99 million, a reduction of 41.1%. The median residual MIP gap at the time limit decreases from 58.9% to 33.0%.

A paired Wilcoxon signed-rank test across the 31 instances confirms that the residual gaps are lower with the strengthened formulation ($p = 1.9 \times 10^{-9}$). The difference between the two median gaps is 25.9 percentage points. These results show that the preprocessing substantially reduces the formulation size and improves the solver’s ability to certify solution quality.

The strengthened MILP is therefore used in the remaining experiments. Four additional arc- and node-elimination rules tested during model development did not yield a measurable improvement and are described in [Appendix C](#).

4.5. Scalability and budget calibration

The small-instance experiments in [Section 4.3](#) provide an implementation check when the optimum is known. We now examine the behavior of the strengthened MILP on the operational instances and select the time limit used in the method comparison.

MILP performance under extended time limits. We solve the strengthened MILP on all 31 operational instances using time limits of 30 minutes, 2 hours, and 6 hours. No instance is solved to proven optimality within these limits. As shown in [Table 7](#), the median residual MIP gap decreases only from 33.0% after 30 minutes to 32.1% after 6 hours. The limited progress over longer runs suggests that proving optimality is difficult mainly because the solver improves the bound slowly.

These results do not imply that the MILP fails to find useful feasible solutions. They show only that it cannot certify their optimality within the tested time limits. [Section 4.6](#) therefore evaluates the quality of the feasible solutions found under a time budget compatible with daily planning.

Table 7: Residual MIP gaps for the strengthened formulation on the 31 operational instances. No instance is solved to proven optimality within the tested time limits.

Time limit	Median gap	Mean gap	Instances solved to optimality
30 minutes	33.04%	35.33%	0/31
2 hours	32.24%	34.66%	0/31
6 hours	32.12%	34.25%	0/31

Selection of the comparison budget. Each run in the method comparison is limited to 600 seconds per instance. This time limit was selected as compatible with the daily planning process. To check whether it stops the heuristic searches too early, we also run the GA and ALNS for 1,800 seconds on every operational instance using 10 random seeds.

For each run, we calculate the share of the total improvement observed by 1,800 seconds that has already been achieved after 600 seconds. Across the 310 runs of each method, the median share is 99.3% for ALNS, with values ranging from 90% to 100%. For the GA, the median is 90.2%, with values ranging from 80% to 100%.

ALNS therefore shows little additional improvement between 600 and 1,800 seconds. The GA continues to improve in some runs, although most of the improvement observed over 1,800 seconds is already achieved within the first 600 seconds. We consequently use a 600-second limit for each run in the operational comparison.

4.6. Comparative performance at the operational budget

We compare the solution approaches on the 21 evaluation instances using the 600-second time limit selected in Section 4.5. The comparison includes the strengthened MILP, the Baseline MILP without the Ω exclusion, the Hexaly formulation, the GA, and the ALNS matheuristic. The Baseline MILP is included to assess the effect of the preprocessing and is not considered a separate planning approach.

All solutions are evaluated using the objective in Equation (1). Each MILP variant is run once per instance. Hexaly, the GA, and the ALNS are each run under 30 independent random seeds. For these randomized methods, the median objective value represents the performance of a typical run, while the best objective value across seeds is used in the repeated-search comparison.

For each instance, the best-known feasible value is the lowest objective value obtained by any method under any seed. The performance profile in Figure 2 is based on the best result obtained by each randomized method and the single result obtained by each MILP variant. For a threshold τ , the profile reports the proportion of instances for which a method is within τ of the best-known feasible value.

At the 10% threshold, Hexaly, the strengthened MILP, and the Baseline MILP each cover at least 95% of the evaluation instances. The corresponding proportion is 90% for ALNS and 57% for the GA. The GA requires a threshold of 30% to cover at least 95% of the instances.

Hexaly attains the best-known feasible value on 12 of the 21 instances. The strengthened MILP does so on 6 instances and lies within 5% of the best-known value on 81% of the instances. Under the repeated-search protocol, Hexaly therefore finds the best-known value most frequently, while the strengthened MILP remains close to it on most instances.

For each randomized method, we summarize the 30 seeds per instance in two ways. The best-of-seeds value takes the lowest objective attained on that instance across all seeds, reflecting performance under repeated search; the median-of-seeds value takes the median objective across seeds, reflecting the performance of a typical single run. The deterministic MILP variants are run once per instance, so the two coincide. For each method and instance, we compute the deviation of these values from the best-known feasible value, and report the median and mean of these deviations across the 21 instances.

Under best-of-seeds, Hexaly has a median deviation of 0.0% and a mean of 0.75%, and the strengthened MILP a median of 0.4% and a mean of 2.8%; the corresponding values are 3.7% and 4.5% for the Baseline MILP, 2.7% and 5.3% for ALNS, and 6.2% and 10.0% for the GA. Under median-of-seeds, which reflects a typical single run, the randomized methods degrade markedly: Hexaly's median deviation rises to 3.1%, ALNS to 8.5%, and the GA to 10.3%, while the deterministic MILP variants are unchanged. A single run of the GA or ALNS is therefore substantially

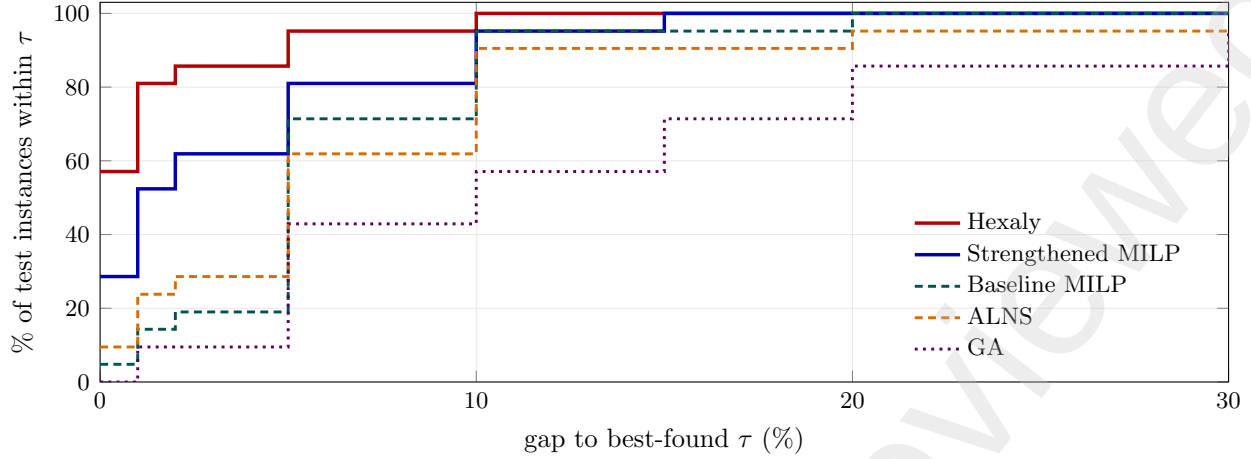


Figure 2: Performance profile on the 21 test instances. For each gap threshold τ , each curve gives the fraction of instances on which the method’s solution lies within τ of the best objective found by any method. The reference pools all 30 seeds of every stochastic method. Curves higher and further to the left indicate stronger performance. Hexaly and the strengthened MILP form the leading tier, each within 10% of the best on at least 95% of instances, while the GA is the weakest, requiring a 30% threshold to reach comparable coverage.

less reliable than repeated search suggests, which motivates the repeated-seed protocol adopted throughout.

We assess statistical significance under the best-of-seeds comparison. A Friedman test rejects the null hypothesis of equal performance across the five methods ($\chi^2 = 37.8$, $p = 3.4 \times 10^{-7}$). Pairwise Wilcoxon signed-rank tests with Holm correction detect no significant difference between Hexaly and the strengthened MILP. Both methods significantly outperform the GA. Hexaly also significantly outperforms the Baseline MILP and ALNS, whereas the differences between the strengthened MILP and these two methods are not significant after correction, with adjusted p -values of 0.55 and 0.42, respectively.

Repeating the analysis on the median-of-seeds values, which reflect a typical single run, yields a stronger result ($\chi^2 = 48.8$, $p = 4.3 \times 10^{-9}$) and a clearer two-tier separation. Hexaly, the strengthened MILP, and the Baseline MILP each significantly outperform both the GA and the ALNS, and are mutually indistinguishable. Best-of-seeds is the more conservative comparison for the randomized methods; their disadvantage relative to the exact and solver-native formulations is more pronounced under typical-run performance. According to these two tests, Hexaly and the strengthened MILP thus provide the strongest and most consistent performance.

The strengthened and Baseline MILP formulations produce similar objective values within the 600-second limit. However, Section 4.4 shows that the Ω exclusion reduces the median residual MIP gap from 58.9% to 33.0%. Taken together, these results suggest that the main benefit of the preprocessing is improved solution certification, rather than a systematic improvement in the incumbents found within 600 seconds.

Overall, the MILP produces feasible solutions that are close to the best-known values despite its large residual MIP gaps. This suggests that the main computational difficulty lies in proving solution quality rather than in finding strong incumbents. Hexaly contributes most frequently to the best-known feasible reference under repeated runs. The GA and ALNS provide additional independent searches and help assess whether better feasible solutions can be found outside the two solver-based formulations.

5. Managerial insights

The computational study shows that the model can generate strong feasible schedules within the time available for daily planning. We now use it to examine three managerial questions. First, what operational cost is associated with improving continuity and appointment-time stability? Second, do the most recently communicated appointment time and the longer-term baseline routine capture different planning concerns? Third, is the observed coverage shortfall mainly explained by the objective weights, by individually infeasible visits, or by the alignment of available capacity with demand?

For the first two analyses, we fix the set of appointments served in the strengthened-MILP solution obtained under the operational weights. More precisely, the variables z_i are fixed, while caregiver assignments, routes, and service times remain free to change. This prevents small changes in coverage from dominating changes in travel, waiting, overtime, continuity, and temporal stability. The resulting curves therefore reflect reallocations among these criteria for a common served set.

We vary the relevant objective weights using weighted-sum scalarization. This method recovers supported efficient solutions but may omit unsupported points on a non-convex frontier (Geoffrion, 1968; Ehrgott, 2005). On the two small instances, an ε -constraint sweep found no additional efficient points over the tested range. Details are provided in Appendix D. For the operational instances, where the exact frontier is unavailable, the resulting curves are interpreted as empirical trade-off paths.

These experiments use the MILP formulation and do not involve heuristic calibration. The representative instances are therefore selected from the full 31-day period rather than only from the held-out evaluation set.

Figure 3 shows one representative schedule produced by the strengthened MILP. It illustrates the assignment and sequencing of visits, caregiver travel, scheduled breaks, available overtime, and examples of short- and long-term temporal deviations.

5.1. The efficiency-patient-experience trade-off

We divide the secondary objective terms into two groups. The *efficiency* group contains travel time, waiting time, and overtime. The *patient-experience* group contains short-term temporal deviation Δ_i^t , long-term temporal deviation Δ_i^o , continuity with the previous caregiver δ_i^{prev} , caregiver-preference costs, and household coordination. Holding the served set fixed, we multiply the weights of the patient-experience group by a scalar λ , ranging from 0 to 500, while keeping the original relative weights within each group unchanged. The operational weighting corresponds to $\lambda = 1$. For each value of λ , we record the resulting weighted efficiency cost and weighted patient-experience penalty.

Figure 4 shows the resulting trade-off paths for two small instances, for which optimal solutions are available, and three operational instances. Each curve is normalized relative to its own $\lambda = 0$ solution to make the shapes comparable across instances of different size. The reported percentages therefore describe changes in the two weighted aggregates, not equal percentage changes in every underlying component.

Across the five instances, the largest reductions in the patient-experience penalty generally occur in the early part of the curve. On *Small test 2*, increasing λ from 0 to 0.5 reduces the patient-experience penalty by 12% while increasing the efficiency cost by only 0.4%. On the operational instance of 2025-01-26, an early increase in λ reduces the patient-experience penalty by 23% with almost no change in the efficiency cost. On 2025-01-04, a 12% reduction is associated with a 4% increase in efficiency cost. Further reductions generally require progressively larger increases in travel, waiting, or overtime.

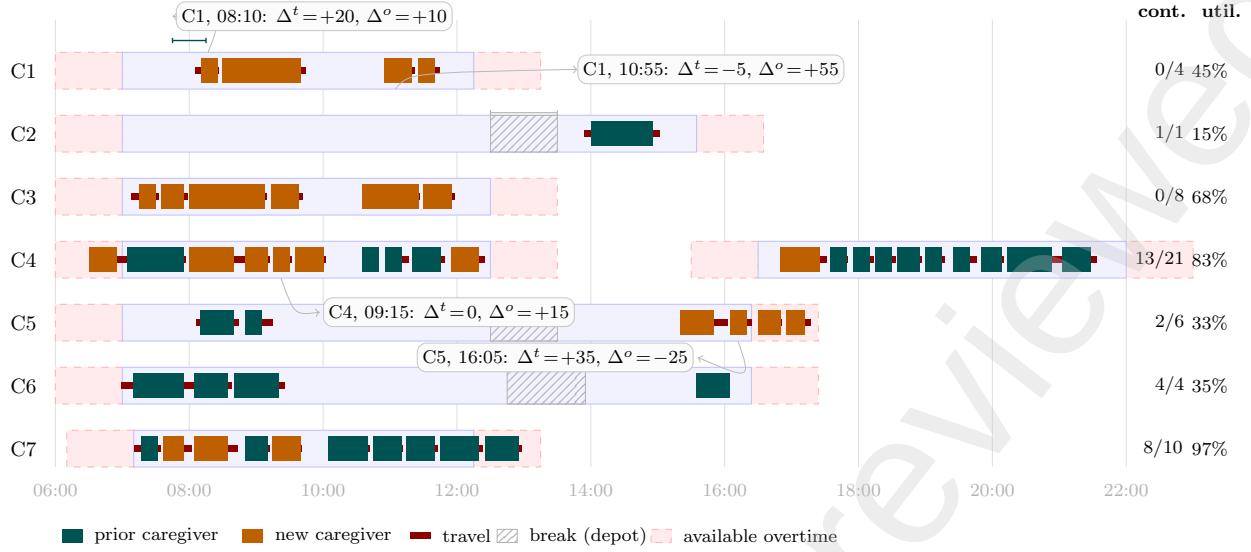


Figure 3: A representative optimized schedule for a single day (2025-01-26), produced by the strengthened MILP. Each row is a caregiver; blue bands are contracted shift windows and the light-red bands are the available overtime (the permitted early-start and late-end tolerance). Coloured blocks are visits (teal: previously assigned caregiver retained; orange: new caregiver); short red segments are travel between locations. Hatched slots are scheduled breaks, taken at the depot within the permitted break window. Caregiver C4 works a split shift. The right-hand columns report continuity-preserving visits and travel-inclusive utilization per caregiver. Callouts report, for several visits, the short-term (Δ^t) and long-term deviation (Δ^o).

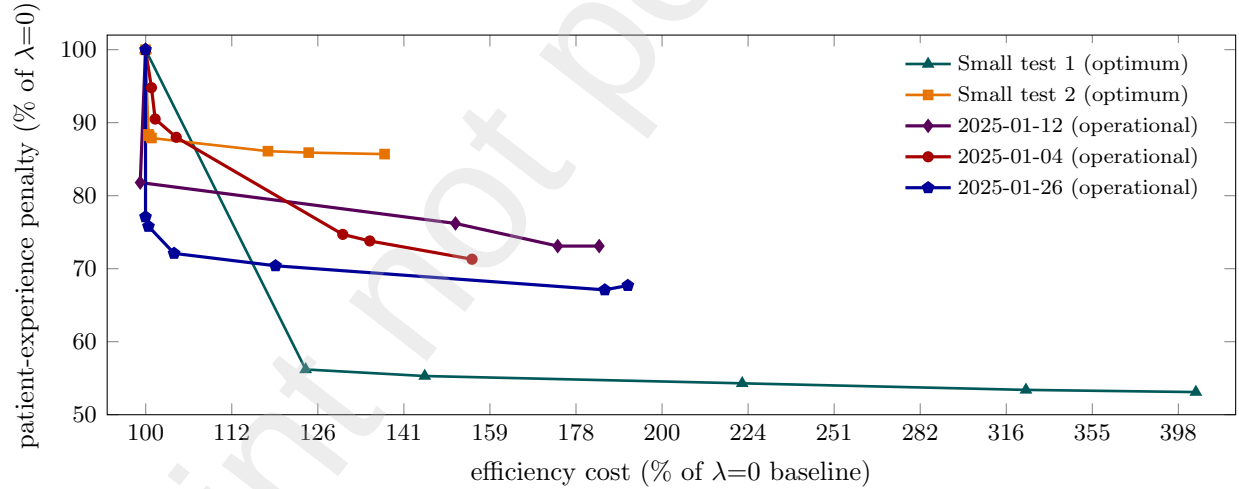


Figure 4: Normalized efficiency-patient-experience trade-off paths for two small instances (for which optimal solutions are available) and three operational instances. Each curve is normalized to its own $\lambda = 0$ solution, so the axes show relative changes in the two weighted aggregates rather than absolute costs. As λ increases, the patient-experience penalty falls most steeply in the early part of each path, with efficiency cost rising more sharply only at larger λ ; the strength of this pattern varies across the instances shown.

These patterns suggest that, for the instances examined, moderate increases in the emphasis placed on patient-experience criteria can yield meaningful improvements at limited operational cost. Under the operational weighting, the schedules still leave room for improvement. Across served visits, 86.1% have a previous-caregiver reference, while only 27.7% retain that caregiver. The mean short- and long-term temporal deviations are 24.0 and 35.2 minutes, respectively, as

reported in Table 8. The trade-off paths indicate that part of this improvement may be obtained before travel, waiting, and overtime begin to rise more sharply. This interpretation applies to the instances examined rather than establishing a common frontier for all operating days.

5.2. Non-redundancy of the two temporal horizons

The model distinguishes two temporal references. The short-term deviation Δ_i^t measures the distance from the appointment time most recently communicated to the patient, while the long-term deviation Δ_i^o measures the distance from the baseline routine established earlier in the care episode. The two references coincide when the most recently communicated time matches the baseline. When they differ, the two terms capture separate scheduling concerns: preserving a recent commitment and limiting longer-term drift from the patient's routine.

The distinction is relevant for most visits in the data. The two reference times differ for 3,918 of the 4,326 visits, or 91%. The daily share ranges from 87% to 94%, and the median absolute difference between the references is 30 minutes. Using only one reference would therefore omit information about either the most recent schedule commitment or the longer-term routine.

The weight-sensitivity experiment examines how the two deviations respond to changes in their respective weights on three operational instances, with the served set fixed.

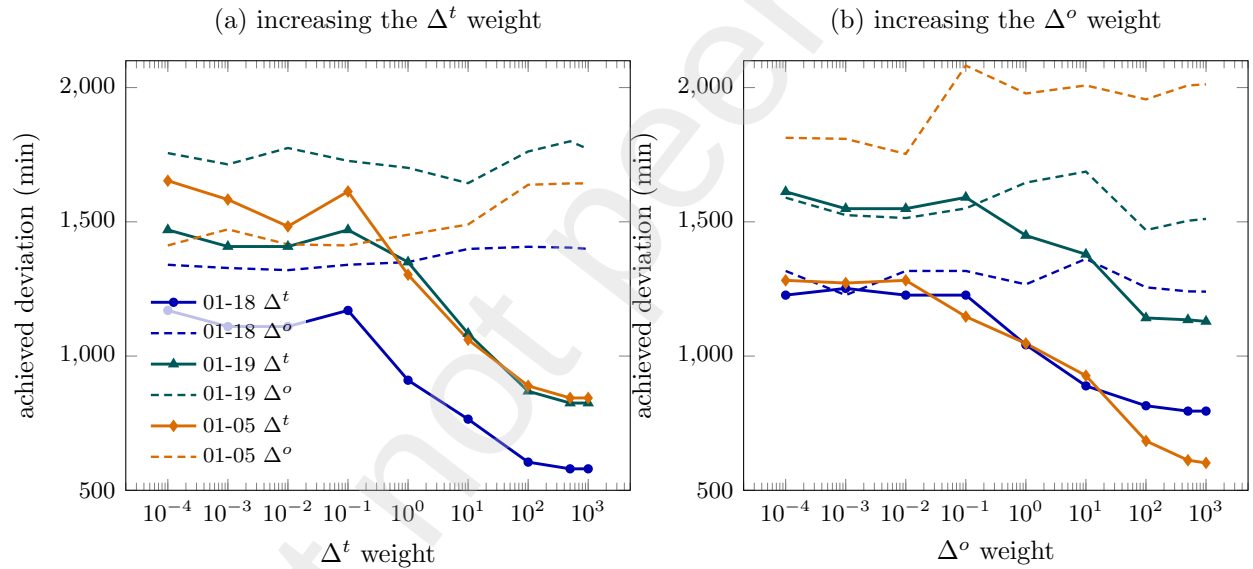


Figure 5: Non-redundancy of the two temporal horizons across three operational instances, with the served set fixed at the operational solution so that only the timing reallocation varies. Each panel plots the aggregate deviation summed over the served visits, $\sum_i \Delta_i^t$ and $\sum_i \Delta_i^o$, as the corresponding weight is increased. The solid curves show the deviation driven down by its own weight, and the dashed curves the other deviation. In panel (a), increasing the short-horizon weight reduces $\sum_i \Delta_i^t$ by 44–50% while $\sum_i \Delta_i^o$ does not follow and rises in two instances. In panel (b), increasing the long-horizon weight reduces $\sum_i \Delta_i^o$ by 30–53% while $\sum_i \Delta_i^t$ stays high. Neither horizon is controlled by the weight on the other, so the two terms are not interchangeable.

In panel (a), increasing the short-term weight while setting the long-term weight to zero reduces the aggregate short-term deviation by 44% to 50%. The long-term deviation does not decrease with it and rises in two of the three instances. Panel (b) shows the corresponding result for the long-term weight. The aggregate long-term deviation decreases by 30% to 53%, while the short-term deviation remains comparatively high.

These results show that improving adherence to the most recently communicated times does not necessarily improve adherence to the baseline routines. The two terms should therefore be

retained separately when both forms of stability matter. Their relative weights allow planners to express how much importance is placed on recent schedule commitments and on longer-term routine stability.

5.3. The coverage shortfall

Table 8 summarizes the schedules produced by the strengthened MILP over the 31 operational days.

Table 8: Operational profile of the schedules produced by the strengthened MILP, averaged over the 31 operational days. Service-and-travel utilization is expressed as a fraction of contracted shift time. Overtime usage is the share of the permitted early-start and late-end tolerance used by the schedules.

Indicator	Mean
Requested visits per day	161.1
Preprocessing-infeasible visits per day	4.6
Served visits per day	73.1
<i>Coverage</i>	
Model, by count / by service minutes	45.4% / 50.0%
Model, excluding preprocessing-infeasible visits	46.7% / 51.7%
Recorded realized schedules, by count / by service minutes	50.0% / 50.0%
Visits with a previous-caregiver reference	86.1%
Continuity with the previous caregiver	27.7%
Short-term deviation Δ^t , mean	24.0 min
within 30 minutes	75.0%
Long-term deviation Δ^o , mean	35.2 min
within 30 minutes	61.5%
Service-and-travel utilization	70.0%
Overtime tolerance used	16.6%

The strengthened-MILP schedules serve 45.4% of requested visits and 50.0% of requested service minutes. Excluding the 4.6 visits per day identified during preprocessing as infeasible increases count-based coverage to 46.7%.⁴ Individually infeasible visits therefore explain only a small part of the observed shortfall.

Service and travel account for an average of 70.0% of contracted shift time. The remaining time should not be interpreted entirely as available capacity, since it may include mandatory breaks, waiting, and time fragments that cannot be matched with feasible visits. The coexistence of unserved demand and unused contracted time nevertheless indicates that total caregiver hours alone do not determine usable capacity.

We first examine whether the coverage shortfall is sensitive to the weight assigned to unserved demand. Four operational instances are re-solved while increasing α_1 from its operational value of 250 to 500,000, with all other weights unchanged and assignments, routing, and timing free to change. Figure 6 shows that the number of served visits changes little over this range. It remains at 51 on 2025-01-12, varies between 53 and 55 on 2025-01-26, and between 108 and 114 on 2025-01-20.

⁴These visits have no compatible caregiver shift under the qualification, timing, and break checks described in Section 4.4.

On 2025-01-04, it increases from 58 to 63 and then remains within this range as the weight increases further.

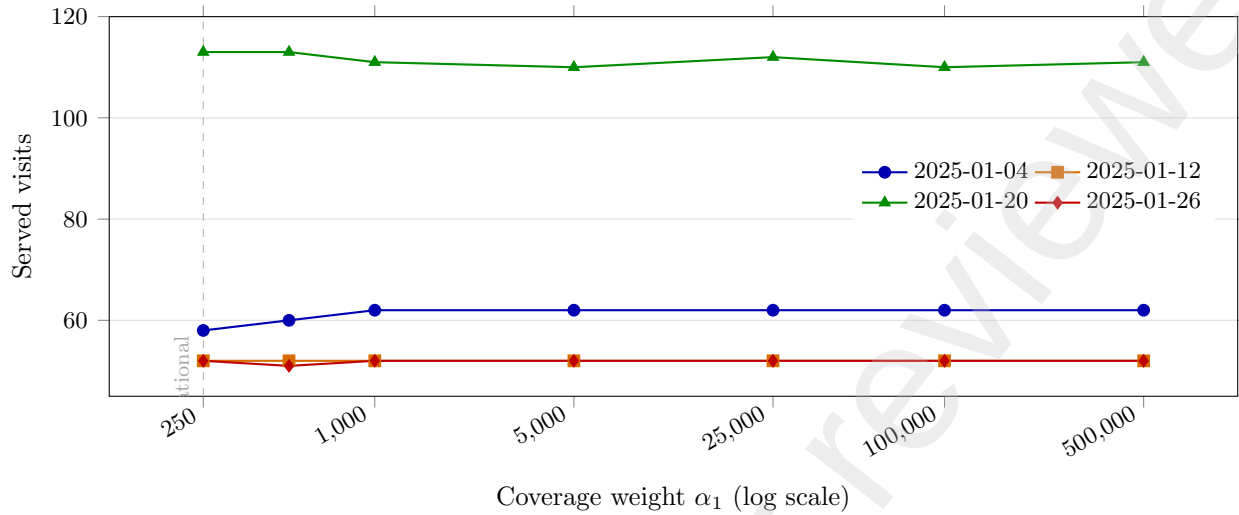


Figure 6: Served visits as a function of the coverage weight α_1 on four representative instances, with all other weights fixed and routing and timing free. The number of served visits changes little as α_1 increases across more than three orders of magnitude.

On 2025-01-04, the increase in coverage is accompanied by an increase in overtime from 138 to 273 minutes. On the other three instances, increasing the coverage weight produces no consistent increase in either coverage or overtime. These results indicate that the operational coverage weight is not the main reason that visits remain unserved on the instances examined.

We next give coverage lexicographic priority on all 31 instances. Relative to the full weighted objective, this produces an average of 0.9 additional visits per day, corresponding to a 1.2% increase. The mean difference in the coverage-objective value is 45.5 min^{1.1}. Some instances show a small negative difference because neither solve reaches proven optimality within the available time.

Finally, we optimize coverage alone for 12 hours per instance using both the strengthened MILP and Hexaly. Average count-based coverage increases from 45.4% to 46.6%, or approximately two additional visits per day. The two methods also produce similar coverage levels over the month. Because neither method proves optimality on the operational instances, these experiments do not establish an exact coverage ceiling. Their consistency across different weights, objective structures, time limits, and solution methods nevertheless provides strong evidence that the operational objective weights are not the main cause of the shortfall.

For context, the realized schedules recorded by the data-providing HHC organization serve approximately 50% of recorded visits by both count and service minutes. The model serves a smaller share by count but the same share of service minutes, indicating that it selects longer appointments more frequently. This behavior is consistent with the superlinear duration term in the unserved-demand penalty. The comparison is descriptive, since realized operations may reflect information or decisions that are not represented in the model.

5.4. Deployment and scope

The computational comparison identifies Hexaly and the strengthened MILP as the two strongest solution approaches in this study. Under the repeated-search protocol, with a 600-second budget per run, Hexaly contributes the best-known feasible solution most frequently. The strengthened

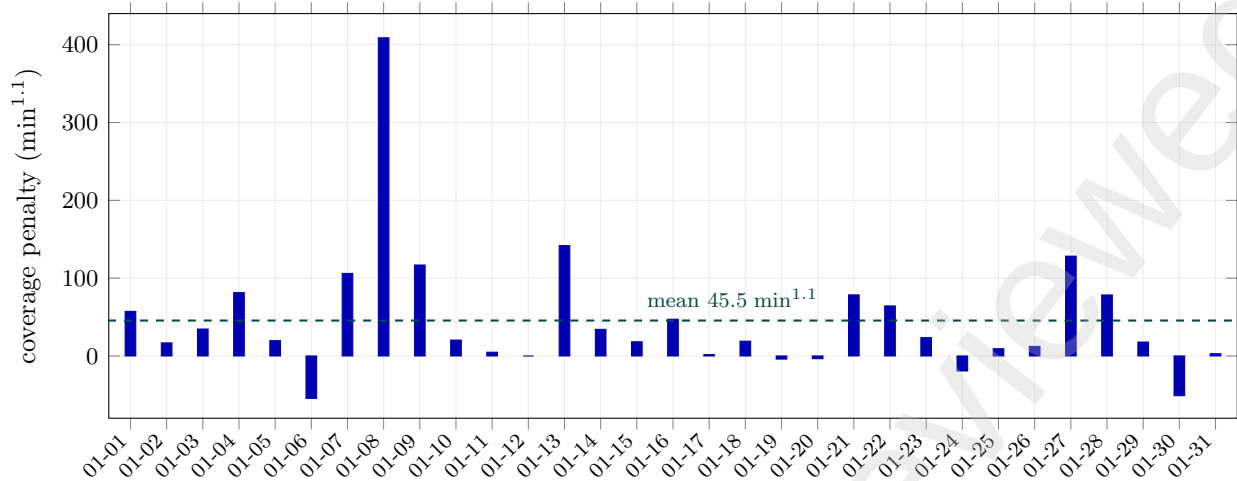


Figure 7: Coverage cost of the secondary objectives, measured in the coverage-objective unit ($\text{min}^{1.1}$). Each bar is the additional coverage penalty incurred by the full weighted solve on a given day relative to the lexicographic coverage-only solve; positive values indicate coverage sacrificed to the efficiency and temporal-stability criteria. The mean difference is $45.5 \text{ min}^{1.1}$ per day, equivalent to 0.9 additional visits served on average, so the inclusion of the secondary criteria imposes a negligible coverage cost. Small negative values reflect that, at operational scale, both solves terminate at the time limit rather than at proven optima. The instance of 2025-01-08 is the largest single-day cost at $409 \text{ min}^{1.1}$ (six additional visits).

MILP attains a median deviation of 0.4% from the best-known feasible value, while also providing a transparent MILP representation and conventional solver bounds. At the evaluation size of 21 instances, the paired tests do not distinguish Hexaly from the strengthened MILP; this indicates that the available data do not separate the two methods on their performance.

These results make both approaches relevant candidates for further deployment testing. Because the comparison includes repeated stochastic runs, it does not by itself determine typical single-run performance. A final implementation choice would also depend on integration with the existing planning software, maintenance requirements, licensing, and the information required by planners. In the present study, the GA and ALNS primarily serve as independent feasible-search benchmarks.

The managerial findings apply to the modeled objective, one month of data, and the operational setting represented in the study. They do not cover preferences, contractual rules, or patient-specific considerations that are absent from the data. The weights used in practice must therefore reflect both the modeled trade-offs and considerations that the formulation does not capture.

Break scheduling and synchronization were introduced through controlled modifications because these features were not directly represented in the raw records. They are not used to support conclusions specific to the data-providing HHC organization. Repeating the analyses on the unmodified instances leaves coverage essentially unchanged and shifts the operational point estimates in a predictable direction, without altering any of the managerial conclusions; the full comparison is reported in [Appendix E](#).

6. Conclusion

This paper develops an integrated model for daily home healthcare routing and scheduling. The model combines caregiver assignment, routing, synchronized visits, break scheduling, patient-caregiver preferences, continuity with previously assigned caregivers, household coordination, and appointment-time stability. A central feature is the distinction between two temporal references:

the appointment time most recently communicated to the patient and the baseline routine established earlier in the care episode. This allows recent schedule commitments and longer-term routine stability to be represented separately.

The computational results show that the proposed formulations can generate strong feasible schedules within the time available for daily planning. The strengthened MILP attains a median deviation of 0.4% from the best-known feasible value obtained across the MILP, Hexaly, GA, and ALNS searches. Under the repeated-search comparison, Hexaly contributes the best-known feasible solution most frequently, while the paired tests do not distinguish it from the strengthened MILP at the evaluation size. The preprocessing based on infeasible appointment–shift pairs reduces the mean number of presolved nonzeros by 41.1% and the median residual MILP gap from 58.9% to 33.0%, without excluding feasible schedules.

The managerial experiments yield three main findings. First, increasing the priority assigned to coverage produces little additional coverage. Even when coverage is optimized alone for 12 hours, average coverage increases only from 45.4% to 46.6%. Together with the unused contracted time observed in the schedules, this suggests that the coverage shortfall is driven mainly by the temporal, spatial, and qualification alignment of capacity with demand, rather than by the objective weights alone.

Second, on the instances examined, moderate increases in the emphasis placed on continuity and temporal stability improve these criteria at limited additional travel, waiting, and overtime cost. Larger improvements become progressively more expensive. Third, the short- and long-term temporal deviations respond differently to their respective weights. This supports representing the appointment time most recently communicated to the patient and the longer-term baseline routine as separate components of the objective.

The empirical analysis is based on one HHC organization and one month of operational data. Break scheduling and synchronized visits were introduced through controlled modifications because these features were not directly represented in the raw records. They are therefore not used to draw conclusions specific to the data-providing organization. The main results were also checked on the unmodified instances and remained similar. The model further treats each day as a static and deterministic planning problem and does not represent uncertainty in demand, cancellations, travel times, or caregiver availability.

Future research could extend the model to rolling-horizon, stochastic, and multi-period settings. Testing it on data from additional HHC organizations would help determine whether the observed coverage and trade-off patterns also arise in other operational environments. Richer preference data could also support a more detailed calibration of the weights placed on caregiver assignment, continuity, temporal stability, and operational efficiency.

Overall, the study provides a practical optimization approach for constructing daily HHC schedules and for examining how coverage, operational efficiency, continuity, and appointment-time stability interact.

References

- de Aguiar, A.R.P., Ramos, T.R.P., Gomes, M.I., 2023. Home care routing and scheduling problem with teams' synchronization. *Socio-Economic Planning Sciences* 86, 101503.
- Algendi, A., Urrutia, S., Hvattum, L.M., Helgheim, B.I., 2025. Home Healthcare Staffing, Routing, and Scheduling Problem With Multiple Shifts and Emergency Considerations. *Networks* 85, 223–242.

- Alonso, T.H., Grampp, M., 2025. Ageing Switzerland: Pathways to a resilient healthcare system — Deloitte Switzerland. Technical Report. Deloitte.
- Bazirha, M., Kadrani, A., Benmansour, R., 2023. Stochastic home health care routing and scheduling problem with multiple synchronized services. *Annals of Operations Research* 320, 573–601.
- Benoist, T., Estellon, B., Gardi, F., Megel, R., Nouioua, K., 2011. LocalSolver 1.x: A black-box local-search solver for 0-1 programming. *4OR* 9, 299–316.
- Bonomi, V., Côté, J.F., Mansini, R., Zanotti, R., 2025. Dynamically dealing with requests in a stochastic multi-period home healthcare problem with consistency constraints. *International Journal of Production Research* 63, 4137–4162.
- Braekers, K., Hartl, R.F., Parragh, S.N., Tricoire, F., 2016. A bi-objective home care scheduling problem: Analyzing the trade-off between costs and client inconvenience. *European Journal of Operational Research* 248, 428–443.
- Bredström, D., Rönnqvist, M., 2008. Combined vehicle routing and scheduling with temporal precedence and synchronization constraints. *European Journal of Operational Research* 191, 19–31.
- Cappanera, P., Scutellà, M.G., 2022. Addressing consistency and demand uncertainty in the Home Care planning problem. *Flexible Services and Manufacturing Journal* 34, 1–39.
- Carello, G., Lanzarone, E., Mattia, S., 2018. Trade-off between stakeholders' goals in the home care nurse-to-patient assignment problem. *Operations Research for Health Care* 16, 29–40.
- Ceschia, S., Di Gaspero, L., Rosati, R.M., Schaerf, A., 2026. Multi-neighborhood simulated annealing for the home healthcare routing and scheduling problem. *International Transactions in Operational Research* 33, 38–67.
- Chirayil Chandrasekharan, R., Smet, P., Wauters, T., 2021. An automatic constructive matheuristic for the shift minimization personnel task scheduling problem. *Journal of Heuristics* 27, 205–227.
- Cissé, M., Yalçındağ, S., Kergosien, Y., Şahin, E., Lenté, C., Matta, A., 2017. OR problems related to Home Health Care: A review of relevant routing and scheduling problems. *Operations Research for Health Care* 13-14, 1–22.
- Coelho, L.C., Gagliardi, J.P., Renaud, J., Ruiz, A., 2016. Solving the vehicle routing problem with lunch break arising in the furniture delivery industry. *Journal of the Operational Research Society* 67, 743–751.
- Deb, K., Agrawal, R., 1995. Simulated Binary Crossover for Continuous Search Space. *Complex Systems* 9, 115–148.
- Deb, K., Goyal, M., 1996. A combined genetic adaptive search (GeneAS) for engineering design. *Computer Science and Informatics* 26, 30–45.
- Decerle, J., Grunder, O., Hajjam El Hassani, A., Barakat, O., 2018. A memetic algorithm for a home health care routing and scheduling problem. *Operations Research for Health Care* 16, 59–71.

- Decerle, J., Grunder, O., Hajjam El Hassani, A., Barakat, O., 2019. A hybrid memetic-ant colony optimization algorithm for the home health care problem with time window, synchronization and working time balancing. *Swarm and Evolutionary Computation* 46, 171–183.
- Delaet, A., Ramaekers, K., Hirsch, P., Molenbruch, Y., Braekers, K., 2024. A matheuristic for integrated medium-term home healthcare planning. *European Journal of Operational Research* 319, 543–556.
- Demirbilek, M., Branke, J., Strauss, A., 2019. Dynamically accepting and scheduling patients for home healthcare. *Health Care Management Science* 22, 140–155.
- Ehrgott, M., 2005. *Multicriteria optimization*. 2 ed., Springer Berlin Heidelberg.
- Euchi, J., Masmoudi, M., Siarry, P., 2022. Home health care routing and scheduling problems: a literature review. *4OR* 20, 351–389.
- Eurostat, 2020. Ageing Europe-statistics on population developments Statistics Explained. Technical Report.
- Fathollahi-Fard, A.M., Ahmadi, A., Karimi, B., 2021. Multi-Objective Optimization of Home Healthcare with Working-Time Balancing and Care Continuity. *Sustainability* 2021, Vol. 13, Page 12431 13, 12431.
- Federal Statistical Office (FSO), 2025. Population projections. Technical Report.
- Ferone, D., Festa, P., Fugaro, S., Pastore, T., 2025. Hybridizing a matheuristic with ALNS for the optimal collection and delivery of medical specimens. *International Transactions in Operational Research* 32, 90–116.
- Fikar, C., Hirsch, P., 2017. Home health care routing and scheduling: A review. *Computers & Operations Research* 77, 86–95.
- Fu, Y., Dong, J., Gao, K., Suganthan, P.N., Huang, M., Zhu, L., 2025. A review on metaheuristics for solving home health care routing and scheduling problems. *Applied Soft Computing* 182, 113560.
- Gardi, F., Benoist, T., Darlay, J., Estellon, B., Megel, R., 2014. *Mathematical Programming Solver Based on Local Search - Frédéric Gardi, Thierry Benoist, Julien Darlay, Bertrand Estellon, Romain Megel - Google Books*. ISTE/Wiley.
- Geoffrion, A.M., 1968. Proper efficiency and the theory of vector maximization. *Journal of Mathematical Analysis and Applications* 22, 618–630.
- Goel, A., Vidal, T., 2013. Hours of Service Regulations in Road Freight Transport: An Optimization-Based International Assessment. *Transportation science* 48, 391–412.
- Grenouilleau, F., Legrain, A., Lahrichi, N., Rousseau, L.M., 2019. A set partitioning heuristic for the home health care routing and scheduling problem. *European Journal of Operational Research* 275, 295–303.
- Grieco, L., Utley, M., Crowe, S., 2021. Operational research applied to decisions in home health care: A systematic literature review. *Journal of the Operational Research Society* 72, 1960–1991.

- Groër, C., Golden, B., Wasil, E., 2008. The Consistent Vehicle Routing Problem. *Manufacturing & Service Operations Management* 11, 630–643.
- Güven-Kocak, S., Heching, A., Keskinocak, P., Toriello, A., 2024. Continuity of care in home health care scheduling: a rolling horizon approach. *Journal of Scheduling* 27, 375–392.
- Kirkpatrick, S., Gelatt, C.D., Vecchi, M.P., 1983. Optimization by Simulated Annealing. *Science* 220, 671–680.
- Kovacs, A.A., Golden, B.L., Hartl, R.F., Parragh, S.N., 2014a. The Generalized Consistent Vehicle Routing Problem. *Transportation science* 49, 796–816.
- Kovacs, A.A., Golden, B.L., Hartl, R.F., Parragh, S.N., 2014b. Vehicle routing problems in which consistency considerations are important: A survey. *Networks* 64, 192–213.
- Kovacs, A.A., Parragh, S.N., Hartl, R.F., 2015. The multi-objective generalized consistent vehicle routing problem. *European Journal of Operational Research* 247, 441–458.
- Kummer, A.F., de Araújo, O.C., Buriol, L.S., Resende, M.G., 2024. A biased random-key genetic algorithm for the home health care problem. *International Transactions in Operational Research* 31, 1859–1889.
- Liu, D., Perreault, V., Hertz, A., Lodi, A., 2022. A machine learning framework for neighbor generation in metaheuristic search .
- Liu, W., Dridi, M., Fei, H., El Hassani, A.H., 2021. Hybrid metaheuristics for solving a home health care routing and scheduling problem with time windows, synchronized visits and lunch breaks. *Expert Systems with Applications* 183, 115307.
- López-Ibáñez, M., Dubois-Lacoste, J., Pérez Cáceres, L., Birattari, M., Stützle, T., 2016. The irace package: Iterated racing for automatic algorithm configuration. *Operations Research Perspectives* 3, 43–58.
- Lourengo, H.R., Martin, O.C., Stützle, T., 2003. Iterated Local Search, in: Glover, F., Kochenberger, G.A. (Eds.), *Handbook of Metaheuristics*. Kluwer Academic Publishers, Boston, pp. 320–353.
- Luo, Z., Qin, H., Che, C., Lim, A., 2015. On service consistency in multi-period vehicle routing. *European Journal of Operational Research* 243, 731–744.
- Mancini, S., Gansterer, M., Hartl, R.F., 2021. The collaborative consistent vehicle routing problem with workload balance. *European Journal of Operational Research* 293, 955–965.
- Mankowska, D.S., Meisel, F., Bierwirth, C., 2014. The home health care routing and scheduling problem with interdependent services. *Health Care Management Science* 17, 15–30.
- Masmoudi, M., Euch, J., Siarry, P., 2024. Home healthcare routing and scheduling: operations research approaches and contemporary challenges. *Annals of Operations Research* 2024 343:2 343, 701–751.
- Masmoudi, M., Jarboui, B., Borchani, R., 2023. Efficient metaheuristics for the home (health)-care routing and scheduling problem with time windows and synchronized visits. *Optimization Letters* 17, 2135–2167.

- Momcheva, A., Salvador, F., Bonet, R., Caserta, M., 2025. On-Demand Schedules, Worker Absenteeism and Patient Dissatisfaction in Home Care Services. *Journal of Operations Management* 71, 1090–1114.
- Montemanni, R., Ceschia, S., Schaerf, A., 2025. A compact model for the home healthcare routing and scheduling problem. *EURO Journal on Computational Optimization* 13, 100101.
- Naderi, B., Begen, M.A., Zaric, G.S., Roshanaei, V., 2023. A novel and efficient exact technique for integrated staffing, assignment, routing, and scheduling of home care services under uncertainty. *Omega* 116, 102805.
- Rasmussen, M.S., Justesen, T., Dohn, A., Larsen, J., 2012. The Home Care Crew Scheduling Problem: Preference-based visit clustering and temporal dependencies. *European Journal of Operational Research* 219, 598–610.
- Rest, K.D., Hirsch, P., 2016. Daily scheduling of home health care services using time-dependent public transport. *Flexible Services and Manufacturing Journal* 28, 495–525.
- Røpke, S., Pisinger, D., 2006. An Adaptive Large Neighborhood Search Heuristic for the Pickup and Delivery Problem with Time Windows. *Transportation Science* , 455–472.
- Sarasola, B., Doerner, K.F., 2019. Adaptive large neighborhood search for the vehicle routing problem with synchronization constraints at the delivery location. *Networks* 75, 64–85.
- Shaw, P., 1998. Using constraint programming and local search methods to solve vehicle routing problems. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 1520, 417–431.
- Stavropoulou, F., 2022. The Consistent Vehicle Routing Problem with heterogeneous fleet. *Computers & Operations Research* 140, 105644.
- Tarantilis, C.D., Stavropoulou, F., Repoussis, P.P., 2012. A template-based Tabu Search algorithm for the Consistent Vehicle Routing Problem. *Expert Systems with Applications* 39, 4233–4239.
- Wang, Y., Hu, W., Hao, J.K., Feng, J., 2025. Collaborative scheduling and routing of home healthcare service across multiple communities. *OR Spectrum* 47, 525–563.
- World Health Organization, 2024. Ageing and health.
- Yadav, N., Tanksale, A., 2022. An integrated routing and scheduling problem for home healthcare delivery with limited person-to-person contact. *European Journal of Operational Research* 303, 1100–1125.

Appendices

Appendix A. Solution approach

The integrated model presented in Section 3 is solved exactly with Gurobi and with the Hexaly formulation of Section 3.3. To provide solver-independent validation of the resulting schedules, we additionally implement two heuristic benchmarks: a hybrid GA and an ALNS matheuristic. The two methods share a common solution representation (Appendix A.1) and a common fitness-evaluation procedure with built-in feasibility repair (Appendix A.2), so that they provide independent reference points rather than reflecting a single implementation. The GA (Appendix A.3) is

population-based, maintaining a pool of candidate schedules under crossover, mutation, and repair operators; the ALNS (Appendix A.5) is single-trajectory, perturbing an incumbent through destroy–repair cycles with inner-LP timing re-optimization. This appendix describes both methods in sufficient detail for reproducibility.

Appendix A.1. Solution representation

Let $\nu_i \geq 1$ denote the number of caregivers required to serve appointment $i \in A$. For appointments requiring synchronized service ($\nu_i > 1$), the appointment is duplicated into ν_i identical assignment genes, each allocated to a distinct shift, but all sharing a common service start time. Let $A' = \{(i, l) : i \in A, l = 1, \dots, \nu_i\}$ denote the expanded set of assignment genes, with $|A'| = \sum_{i \in A} \nu_i$. A solution is represented by a chromosome (a, t) consisting of two components:

- Assignment vector $a = (a_{i,l})_{(i,l) \in A'}$, where $a_{i,l} \in K \cup \{\emptyset\}$ denotes the shift assigned to gene (i, l) , or $\emptyset$ if the appointment is unserved.
- Timing vector $t = (t_i)_{i \in A}$, where $t_i \in [\underline{T}_i, \overline{T}_i]$ denotes the planned service start time of appointment i , shared across all ν_i genes. This ensures that all caregivers assigned to appointment i begin service simultaneously.

The chromosome does not explicitly encode visit order. Instead, the route sequence for each shift is induced by sorting its assigned appointments by planned start time. This mirrors the operational structure of HHC planning, where appointment times are the central output communicated to patients and caregivers, and routes follow from the temporal schedule.

Appendix A.2. Fitness evaluation and feasibility repair

The fitness function corresponds to the weighted objective in Equations (1) and (46). Since search operators may generate infeasible solutions, the evaluation includes a deterministic repair procedure that enforces temporal feasibility and computes all objective components consistently. Before traversing the routes, appointments that require multiple caregivers ($\nu_i \geq 2$) are checked for caregiver distinctness; if two assignment genes of the same appointment are assigned to the same caregiver, all copies are unassigned and the appointment is treated as non-assigned.

For each shift $k \in K$, we extract the set of assigned appointments $A_k = \{(i, l) \in A' : a_{i,l} = k\}$ and sort them in non-decreasing order of their planned start times t_i . We then simulate route execution forward in time. Let τ denote the current time at which the caregiver becomes available. Starting from the shift start time, we sequentially compute travel and service start times. The realized start time of appointment i is updated as $t_i \leftarrow \max\{t_i, \tau + \tau_{\text{travel}}, \underline{T}_i\}$, where τ_{travel} is the travel time from the previous location to appointment i . If $t_i > \overline{T}_i$ or the route's end-node time exceeds the shift end time $\overline{L}_k$, the appointment violates its time window and is removed from the shift. Removed appointments are marked unserved ($a_{i,l} \leftarrow \emptyset, \forall l$) and contribute to the uncovered-demand penalty.

Mandatory rest breaks are inserted into each shift's sequence and treated alongside appointments during forward simulation. If a floating break cannot be scheduled within its allowable window, the most recent preceding appointment is removed and the simulation prefix is recomputed.

The forward simulation yields the travel, waiting, overtime, and start-time quantities needed to score all objective components, so that every evaluated chromosome corresponds to a feasible operational schedule. Algorithm 1 summarizes the evaluation procedure.

Algorithm 1: Fitness evaluation with forward simulation and temporal repair

```
1  $f \leftarrow 0$ 
2 foreach appointment  $i \in A$  with  $\nu_i \geq 2$  do
3   if the  $\nu_i$  assigned shifts are not all distinct caregivers then
4      $a_{i,l} \leftarrow \emptyset$  for all  $l = 1, \dots, \nu_i$ 
5 foreach shift  $k \in K$  do
6    $A_k \leftarrow \{(i, l) \in A' : a_{i,l} = k\} \cup B_k$ , sorted by  $t_i$ 
7    $\tau \leftarrow \underline{L}_k$ ,  $v \leftarrow s_k$  (start time and location)
8   foreach item  $i \in A_k$  in sequence do
9     if  $i \in B_k$  then
10      Schedule break and drop preceding appointments if infeasible (Appendix A.2)
11    else
12      Compute realized start time  $t_i$  from travel time and visiting windows (Appendix A.2)
13      if  $t_i > \bar{T}_i$  or  $t_i + \sigma_i > \bar{L}_k$  then
14         $a_{i,l} \leftarrow \emptyset$ , remove  $(i, l)$  from  $A_k$ , continue
15      Accumulate travel and waiting contributions in  $f$ 
16    Update  $\tau$  and  $v$  to reflect completion of  $i$ 
17  Add return-to-end-node travel time to  $\tau$ 
18  if  $\tau > \bar{L}_k$  then
19    Accumulate overtime contribution in  $f$ 
20 Add uncovered-demand, continuity-of-care, household-coordination, and temporal-stability penalties to  $f$ 
```

Appendix A.3. Genetic algorithm

Genetic and memetic algorithms have been widely applied to HHC routing and scheduling owing to their flexibility with heterogeneous constraints (Decerle et al., 2018, 2019; Bazirha et al., 2023; Kummer et al., 2024). Our GA combines feasibility-preserving repair, adaptive mutation control with reheat-and-cool dynamics, and LP-based local search on route timings. Encoding appointment start times directly as genes, rather than deriving them from a permutation, simplifies multi-caregiver synchronization and enables the continuous LP-based timing local search described in Appendix A.3.5.

Appendix A.3.1. Framework

We employ a generational $(\mu + \lambda)$ GA with population size μ and offspring size λ . Initial solutions are generated by sampling each appointment's shift assignment from a weighted distribution over its feasible shifts, with weights proportional to $1/(1 + \zeta_{i,k})$, where $\zeta_{i,k}$ encodes the continuity-of-care preference (lower ζ equals more preferred). Assignments forbidden by historical incompatibility ($\zeta_{i,k} = \infty$) receive negligible weight, and the unassigned option ($\emptyset$) receives a small fixed weight to promote diversity. The timing genes are initialized by sampling uniformly from the appointment time window. Each individual is then repaired and evaluated as described in Appendix A.2.

In each generation, parents are selected by tournament selection. Two offspring are produced

via crossover and mutation, repaired, and inserted into the population using $\mu + \lambda$ elitist survival selection.

Appendix A.3.2. Crossover and mutation operators

Crossover is applied separately to assignment and timing genes. Assignment genes recombine through a biased uniform crossover: when an appointment is assigned in one parent but unassigned in the other, the assigned value is inherited with high probability, reflecting the operational priority on coverage. Timing genes recombine through simulated binary crossover (Deb and Goyal, 1996) and are clipped to the appointment time window. Mutation comprises three components applied to each offspring:

- *assignment mutation*: each gene $a_{i,l}$ is independently re-sampled from its weighted domain with probability p_m^a .
- *timing mutation*: each gene t_i is perturbed with probability p_m^t by polynomial mutation (Deb and Agrawal, 1995), clipped to the time window.
- *anti-unassignment repair*: a bounded number of unassigned genes with at least one feasible shift are re-sampled, biasing offspring toward higher coverage.

The SBX and polynomial-mutation distribution indices (η_c and η_m) are tuned (Table 5); larger values concentrate offspring near the parent values and smaller values widen the spread. All offspring are repaired after mutation.

Appendix A.3.3. Diversity management and restart strategy

To mitigate premature convergence, diversity is maintained through similarity penalties and restarts. For each individual x during survival ranking, the maximum similarity to the 20 individuals immediately above it is computed and added as a penalty to its weighted fitness. Similarity between chromosomes $x = (a^x, t^x)$ and $y = (a^y, t^y)$ is

$$\text{sim}(x, y) = \frac{1}{1 + \frac{d_H(a^x, a^y)}{|A'|} + \frac{d_1(t^x, t^y)}{d_1^{\max}(t^x, t^y)}},$$

where d_H is the Hamming distance on assignment vectors, d_1 the ℓ_1 distance on timing vectors, and $d_1^{\max}(x, y) = \sum_i \max(t_i^x, t_i^y)$ a normalizing bound; both terms lie in $[0, 1]$.

A restart is triggered when the best weighted fitness fails to improve by at least 1% over a sliding window of W generations. The top $\lceil \eta_e \mu \rceil$ individuals (elites, $\eta_e \in (0, 1)$) are retained; the remainder are reseeded from the ranked parent–offspring pool, skipping individuals whose similarity to a retained one exceeds 0.9. Restarts are capped at $R_{\max}$, after which the algorithm terminates.

Appendix A.3.4. Adaptive exploration–exploitation control

The mutation rates p_m^a and p_m^t are coupled to search progress. On each restart they are multiplied by a reheat factor $\rho_{\text{up}} > 1$ (capped at 0.9 and 0.6); on each improvement of the global incumbent they are multiplied by a cooling factor $\rho_{\text{down}} < 1$ (floored at 0.2 and 0.1), concentrating search around promising regions. For the H generations following a restart (the hot phase), the SBX index is halved, polynomial mutation is replaced by a wider creep mutation, and p_c is set to 1, temporarily widening offspring diversity.

Appendix A.3.5. Local search integration

Local search is applied probabilistically (frequency p_{LS}) to unpolished survivors of the $(\mu + \lambda)$ selection, outside hot phases. Given a fixed assignment vector, it re-optimizes the appointment start times to minimize the time-related penalties (waiting, overtime, temporal stability) while preserving the chromosome-induced visit order. To respect synchronization, same-customer non-overlap, and household coordination, shifts are grouped into connected components: two shifts are linked if they share a multi-caregiver appointment ($\nu_i > 1$), a same-customer pair $(i, j) \in A^c$, or a household pair $(i, j) \in A^l$. One timing LP is solved per component (Appendix A.4); since the assignment and visit order are fixed, it is a continuous LP whose solution weakly dominates the original individual on the time-related terms and replaces it directly. The same LP is reused in the ALNS (Appendix A.5.5).

Appendix A.4. Timing LP

Given a fixed assignment vector and the visit order induced by sorting each shift's appointments by start time, the binary variables $x, \hat{x}, y, z, \delta^{\text{prev}}$ of the MILP become constants, leaving only the continuous timing and penalty variables. For each connected component C (Appendix A.3.5), we solve

$$\min \alpha_4 \sum_{(i,j) \in A^l \cap A_C^2} v_{ij} + \alpha_5 \sum_{i \in A_C} \sum_{k \in C} w_{ik} + \alpha_6 \sum_{k \in C} (o_k^s + o_k^e) + \alpha_7 \sum_{i \in A_C} \Delta_i^t + \alpha_8 \sum_{i \in A_C} \Delta_i^o$$

subject to the time-related constraints of the MILP (node time windows (20), travel-induced precedence (16)–(19)), same-customer ordering (24)–(25), household proximity (26)–(27), waiting time (28)–(30), overtime (31), and temporal-stability linearizations (32)–(33), each restricted to component C with the fixed visit order substituted for the ordering binaries. The terms depending only on the fixed assignment (coverage, travel, preference, continuity) are constant and excluded; they are recombined when scoring the polished individual. Because the assignment and order are fixed, this is a continuous LP solved to optimality per component.

Appendix A.5. Adaptive Large Neighborhood Search

ALNS (Røpke and Pisinger, 2006) is a single-trajectory metaheuristic that alternates between destroying a portion of the current solution and repairing it by reinserting the removed elements. The method maintains a portfolio of destroy and repair operators whose selection probabilities adapt to their recent performance. We adopt an ALNS framework augmented with two components specific to the HHC setting: (i) HHC-aware destroy operators that exploit the temporal and spatial structure of caregiver routes, and (ii) an inner LP that re-optimizes service start times for every candidate solution, given the current assignment and visit order. This matheuristic hybridization separates the NP-hard combinatorial decisions, assignment and route ordering, handled by the ALNS, from the polynomially tractable timing decisions, handled by the LP. The resulting method follows recent matheuristic ALNS approaches in routing and scheduling (Sarasola and Doerner, 2019; Chirayil Chandrasekharan et al., 2021; Liu et al., 2022; Ferone et al., 2025).

Appendix A.5.1. Framework

We employ a single-trajectory ALNS over the same chromosome representation as the GA (Appendix A.1). At each iteration a destroy operator unassigns a subset of appointments from the incumbent, and a repair operator reinserts them along with a bounded sample of appointments that are currently unassigned. The repaired candidate is then polished by the component-wise timing LP

and evaluated by the forward simulator (Algorithm 1). Candidates are accepted by the Metropolis criterion of simulated annealing (Kirkpatrick et al., 1983), and the destroy and repair operators are selected with probabilities updated through the reactive scheme of Røpke and Pisinger (2006).

The initial solution is obtained from a randomized greedy sampler that draws each appointment's shift from its continuity-weighted domain and retains the best of five independent draws. The initial solution is polished once by the timing LP before the main loop begins.

Appendix A.5.2. Destroy and repair operators

At each iteration the destroy size q is drawn uniformly between $q_{\min} = \lceil q_{\min}^{\text{frac}} \cdot n_a \rceil$ and $q_{\max} = \lfloor q_{\max}^{\text{frac}} \cdot n_a \rfloor$, where n_a is the number of currently assigned genes. Five destroy operators are used:

- *Random removal*: removes q assigned genes uniformly at random.
- *Worst removal*: removes the q genes with the largest value of the proxy $r_i = \max\{0, t_i - \underline{T}_i\} + \varepsilon_i$, where a late realized start within the time window signals little forward slack; ε_i breaks ties. The proxy avoids a full re-evaluation per candidate.
- *Related (Shaw) removal*: grows a cluster of q mutually related genes (Shaw, 1998). Relatedness combines geographic distance, realized-start-time difference, and a same-shift indicator,

$$R(i, j) = w_d \cdot \tilde{d}(\ell_i, \ell_j) + w_t \cdot |\tilde{t}_i - \tilde{t}_j| + w_s \cdot \mathbf{1}[k_i \neq k_j],$$

with a determinism exponent controlling cluster tightness.

- *Shift removal*: removes up to q genes from a shift drawn among the longest routes, allowing a caregiver to be re-routed completely.
- *Time-block removal*: removes genes whose realized start falls in a randomly placed 90–180 minute window, clearing a time band for the repair to rebuild against the stability and household penalties.

Each repair reinserts the destroyed genes together with a bounded sample (capped at $\max\{2, q\}$) of already-unassigned appointments; the second component lets coverage grow beyond the initial solution. Three repair operators are used:

- *Random insertion*: assigns each gene to a shift sampled from its continuity-weighted domain, providing the diversification used by the escape mechanism.
- *Greedy insertion*: inserts each gene, in random order, at the feasible shift minimizing the full weighted objective.
- *Regret-2 insertion*: inserts genes in decreasing order of the gap between their best and second-best insertion cost, placing single-feasible-shift genes first.

After repair, start times are re-optimized by the timing LP (Appendix A.4) and the objective is computed via Algorithm 1.

Appendix A.5.3. Operator selection and acceptance

Operator selection. Operators are sampled with probability proportional to weights w_i . Each scores σ_1 , σ_2 , or σ_3 for producing a new global best, an accepted improvement, or an accepted worsening move respectively ($\sigma_1 > \sigma_3 > \sigma_2$); unaccepted applications score zero. Every L_{seg} iterations the weights are updated as $w_i \leftarrow (1 - \rho_w) w_i + \rho_w \frac{s_i}{u_i}$, where s_i and u_i are operator i 's cumulative score and usage over the segment. Weights are floored at a small positive value to keep every operator reachable.

Acceptance. . A candidate is accepted under the Metropolis rule: unconditionally if it does not worsen the objective, else with probability $\exp(-\Delta/T)$. The temperature starts at $T_0 = \kappa \cdot |\text{obj}_0|$ and follows a geometric schedule $T \leftarrow \max\{T_{\min}, \rho_T \cdot T\}$.

Appendix A.5.4. Escape mechanism

To recover from stagnation, an escalating perturbation is triggered by the number of consecutive non-improving iterations Θ (Lourenço et al., 2003):

- *Level 1* ($\Theta \geq \Theta_1$): reset the current solution to the global incumbent and reheat to T_0 .
- *Level 2* ($\Theta \geq \Theta_2$): apply a 40% random destroy to the incumbent, random insertion, and one LP polish; reset T to T_0 .
- *Level 3* ($\Theta \geq \Theta_3$): apply a 60% random destroy, random insertion, and one LP polish; reheat to $2T_0$.

The thresholds are nested ($\Theta_1 < \Theta_2 < \Theta_3$). Boolean flags ℓ_1, ℓ_2, ℓ_3 ensure each level fires at most once per cycle, so Θ continues to climb past Levels 1 and 2 toward the next threshold. Only Level 3 resets $\Theta \leftarrow 0$, clears the flags, and increments a counter ξ , opening a new cycle. All flags and Θ also reset on an improvement of the global incumbent. The mechanism plays the role that population reseeding plays in the GA (Appendix A.3.3).

Appendix A.5.5. Timing LP repair

After every destroy–repair cycle, the candidate’s service start times are re-optimized by the same component-wise LP used as local search in the GA. The decomposition into connected components and the LP formulation are described in Appendix A.3.5 and Appendix A.4, respectively.

Unlike the GA, which applies the LP probabilistically and only to non-polished survivors, the ALNS applies it to every candidate. Each candidate is the basis for the next destroy–repair cycle, so the characteristics of the solution derived by the operators depend on accurate timings; we therefore re-optimize timings for every candidate, not only for accepted ones. The LP is skipped when the candidate contains hard time-window violations detected by the forward simulator, in which case the candidate is evaluated with the simulator’s repaired timings and the LP is retried on the next accepted candidate.

Appendix A.6. Termination criterion

Both metaheuristics terminate at a wall-clock limit. The GA additionally terminates when its restart cap $R_{\max}$ is exhausted (Appendix A.3.3). The ALNS additionally terminates when ξ reaches $R_{\max}$ (that is, after $R_{\max}$ Level-3 cycles pass without improving the global incumbent, since ξ resets on every such improvement; see Appendix A.5.4). The best solution found is returned.

Appendix B. Data values

Table A.1 reports the per-day characteristics of every instance used in the computational study. The first two rows are the verification instances of Section 4.3; the remaining 31 rows are the operational days of January 2025 over which the summary statistics of Table 3 are computed.

Table A.1: Per-instance characteristics underlying the summary in Table 3. The first two rows are the verification instances; the remaining 31 rows are the operational days of January 2025 over which the summary statistics are computed.

Instance	$ A $	$ K ^\dagger$	Customers	$ R $	Sync% [‡]	Prev.Caregiver%	Household	Demand/Supply%	TW med (min)	Travel med (min)
Small test 1	12	8	5	32	8.3	66.7	1	14	100	6.6
Small test 2	12	6	4	3	8.3	91.7	1	19	120	11.0
2025-01-01	147	16	97	15	0.7	70.7	0	94	90	4.5
2025-01-02	155	12	102	15	0.6	63.2	2	91	90	4.3
2025-01-03	180	13	125	34	0.6	68.3	2	124	90	4.4
2025-01-04	117	11	83	34	0.9	56.4	1	92	90	4.8
2025-01-05	119	10	87	18	0.8	71.4	1	84	90	4.7
2025-01-06	179	12	125	34	0.6	63.7	2	113	90	4.0
2025-01-07	176	12	117	32	0.6	60.2	1	110	90	4.4
2025-01-08	179	17	116	33	0.6	30.2	2	90	90	4.4
2025-01-09	193	15	131	33	0.5	24.9	2	101	100	4.4
2025-01-10	123	17	76	34	0.8	0.0	2	61	90	3.9
2025-01-11	133	11	90	33	0.8	22.6	1	91	90	4.5
2025-01-12	111	8	74	16	0.9	23.4	1	105	90	4.4
2025-01-13	218	16	154	20	0.5	28.9	2	93	90	4.2
2025-01-14	179	9	111	34	0.6	22.3	1	134	90	4.3
2025-01-15	192	16	121	34	0.5	25.5	2	107	90	4.2
2025-01-16	113	15	70	33	0.9	0.0	2	68	90	3.8
2025-01-17	185	17	127	35	0.5	24.3	2	93	90	4.4
2025-01-18	120	9	82	33	0.8	17.5	1	105	90	4.7
2025-01-19	109	8	74	17	0.9	19.3	1	93	80	4.3
2025-01-20	135	18	83	34	0.7	0.0	2	53	90	3.6
2025-01-21	182	12	118	34	0.5	19.8	1	113	90	4.2
2025-01-22	192	15	132	33	0.5	22.9	2	103	90	4.4
2025-01-23	189	13	125	34	0.5	24.9	2	99	90	4.5
2025-01-24	183	12	129	37	0.5	28.4	2	114	90	4.3
2025-01-25	134	11	93	34	0.7	26.9	1	99	90	4.3
2025-01-26	103	8	69	16	1.0	23.3	1	80	80	4.4
2025-01-27	219	18	148	35	0.5	25.1	3	97	90	4.3
2025-01-28	174	12	112	33	0.6	24.7	1	118	90	4.3
2025-01-29	169	14	114	33	0.6	23.7	2	73	90	4.3
2025-01-30	189	14	127	33	0.5	26.5	2	97	90	4.2
2025-01-31	196	17	141	34	0.5	30.6	2	90	90	4.5

[‡] Reported for the feature-complete instances used in the computational study, *i.e.* after the two controlled modifications.

Appendix C. Additional preprocessing inequalities

Beyond the infeasible appointment–shift exclusion Ω retained in the main formulation (Section 4.4), we evaluated four finer-grained preprocessing inequalities during model development. Each fixes provably unusable variables to zero and, like Ω , preserves the feasible set established in Section 3.2. We state each candidate and then report its effect when added to the lean formulation.

Appendix C.1. Candidate inequalities

Impossible-appointment fixing. An appointment infeasible in every shift, i.e. with $(i, k) \in \Omega$ for all $k \in K$, cannot be served and is fixed out a priori:

$$z_i = 0, \quad \forall i \in A \text{ such that } (i, k) \in \Omega \quad \forall k \in K. \quad (\text{C.1})$$

Infeasible-arc elimination. An arc whose traversal cannot satisfy time-window precedence is removed. Let

$$\Phi = \{(i, j, k) \in N \times N \times K : \underline{T}_i + \sigma_i + \tau_{ijk} > \overline{T}_j\};$$

then

$$x_{ijk} = 0, \quad \forall (i, j, k) \in \Phi. \quad (\text{C.2})$$

Infeasible break-arc elimination. The analogous fixing is applied to break-spanning arcs $\hat{x}_{ijk}$ that violate the same precedence relation:

$$\hat{x}_{ijk} = 0, \quad \forall (i, j, k) \in \Phi. \quad (\text{C.3})$$

Appendix C.2. Computational effect

Adding each inequality individually to the lean formulation, we solved all 31 operating days under a 1,800-second budget and compared the resulting optimality gap against the lean formulation by paired Wilcoxon signed-rank tests with Holm–Bonferroni correction. Table A.2 reports, per configuration, the median gap, the mean presolved nonzero count and its reduction relative to the baseline formulation, and the Holm-corrected p -value.

Table A.2: Effect of adding each finer-grained preprocessing inequality to the lean formulation, over the 31 operating days under a 1,800-second budget. Reported are the median optimality gap, the mean presolved nonzeros with the reduction relative to the baseline formulation, and the Holm-corrected paired Wilcoxon p -value against the lean formulation.

Configuration	Median gap (%)	Presolved nnz	Reduction (%)	p_{Holm}
Baseline (no Ω)	58.90	3,385,125	–	–
Lean (Ω only)	33.04	1,994,626	41.1	–
+ impossible fixing (C.1)	32.68	1,986,660	41.3	0.57
+ infeasible arc (C.2)	33.66	1,939,830	42.7	1.00
+ infeasible break-arc (C.3)	33.04	1,994,626	41.1	1.00

None of the three candidates produced a statistically significant change in the optimality gap (all $p_{\text{Holm}} > 0.5$), and each left the presolved model size essentially unchanged: the mean nonzero reduction relative to the baseline formulation remained within 1.6 percentage of the lean value of 41.1% in every case. The infeasible-arc elimination (C.2) is illustrative: it added on the order of 45,000 constraints per instance yet yielded no measurable tightening, indicating that the solver’s presolve independently recovers these arc- and node-level fixings. The appointment–shift exclusion

Ω retained in the main formulation, by contrast, encodes qualification, travel-aware temporal, and break-conflict reasoning that the general-purpose presolve does not derive, which accounts for its decisive effect (Section 4.4). These finer-grained inequalities are therefore omitted from the final formulation.

Appendix D. Completeness of the weighted-sum frontier

The efficiency-patient-experience frontiers of Section 5.1 are obtained by weighted-sum scalarization, which recovers only supported efficient solutions and may omit unsupported efficient points on a non-convex frontier (Geoffrion, 1968; Ehrgott, 2005). An apparent bend in the trade-off path could therefore reflect an unrecovered region rather than the trade-off itself. We test this on the two small instances, the only ones for which an exact efficient frontier is computable.

For each instance we minimize the patient-experience family subject to a hard bound ε on the efficiency family, stepping ε across the full efficiency range. Because the ε -constraint method recovers unsupported efficient points that weighted-sum scalarization cannot, any omitted point would appear here as a recovered point dominating the weighted-sum frontier.

Figure D.8 reports the result. On both instances the minimized patient-experience penalty descends to a floor that remains flat across the swept range, and every recovered point coincides with the weighted-sum frontier without dominating it. The weighted-sum frontier is therefore complete over the computable range. Because these instances reduce to a single efficient corner, the check verifies the completeness of scalarization rather than reproducing the trade-off shape of Section 5.1, which arises only on the larger operational instances. Where the frontier is computable, scalarization traces it without artifacts; together with the regular single-bend shape of the operational frontiers in Figure 4, this makes a scalarization artifact an unlikely explanation for the reported trade-off shape.

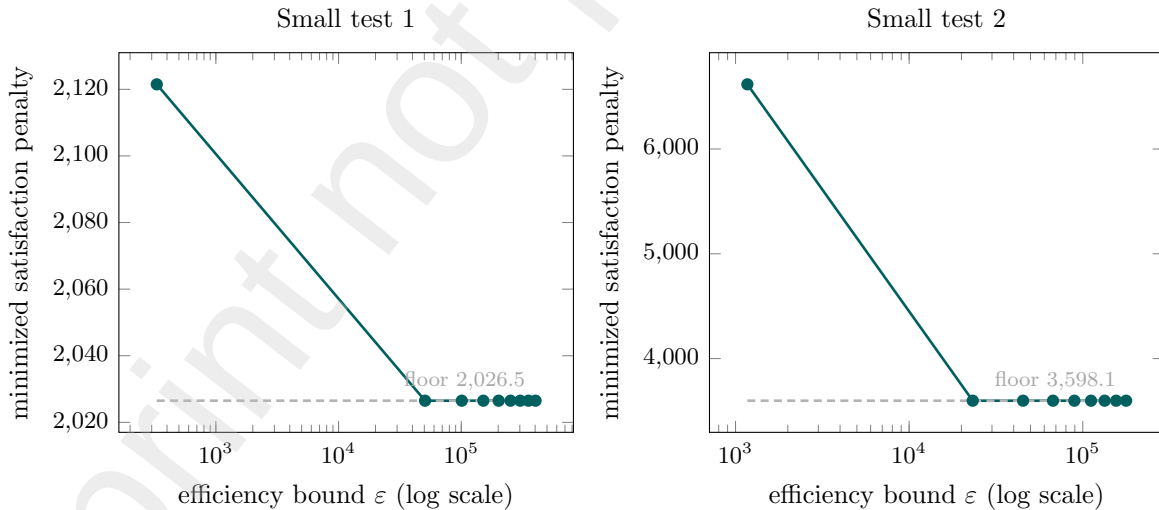


Figure D.8: ε -constraint completeness check on the two instances. Because the ε -constraint method can recover unsupported efficient points that weighted-sum scalarization cannot, their absence here validates that the reported frontier is complete.

Appendix E. Robustness to the controlled modifications

To confirm that the managerial findings do not depend on the controlled modifications of Section 4.1, we repeat the operational solve on the *unmodified* January 2025 instances, in which each caregiver’s half-day shifts are retained separately with fixed break timing and no appointment is synchronized. Table A.3 reports the operational profile in both settings, averaged over the 31 days.

Table A.3: Strengthened-MILP operational profile under the modified and unmodified instances, averaged over the 31 operational days.

Indicator	Modified	Unmodified	Diff.
Served visits per day	73.1	72.3	+0.8
Coverage, by count	45.4%	45.9%	−0.5
Coverage, excl. infeasible	46.7%	47.1%	−0.4
Continuity (same caregiver as prior)	27.7%	31.4%	−3.7
Short-term deviation Δ^t , mean	24.0 min	28.0 min	−4.0
Long-term deviation Δ^o , mean	35.2 min	35.1 min	+0.1
Service-and-travel utilization	70.0%	65.3%	+4.7
Overtime tolerance used	16.6%	11.8%	+4.8

Coverage is essentially unchanged (within 0.5 percentage points), so the shortfall analyzed in Section 5.3 is a property of the instances rather than of the modifications. The remaining differences are modest and move in a consistent direction: combining the half-day shifts and adding the synchronized series uses caregiver time more intensively, raising utilization and used overtime by roughly 5 points, which in turn slightly improves short-term stability while reducing continuity as synchronization prompts a few reassignments. Long-term stability is unaffected. The modifications thus shift the operating point in a predictable way. The three managerial findings of Section 5 are unaffected: the coverage shortfall persists and stays insensitive to the coverage weight, the efficiency–patient–experience trade-off retains its early-steep shape, and the two temporal horizons remain non-redundant. The conclusions drawn from the model therefore do not depend on the controlled modifications.